

A MIN-MAX GAP CHARACTERIZATION OF MINIMAL FOLIATIONS ON THE TORUS

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ABSTRACT. We extend an energy introduced by Mather to the setting of Almgren-Pitts min-max theory and obtain a parametric, higher-dimensional analogue of Mather's variational barrier theory for twist maps and geodesics on tori. We use this energy to establish several criteria for the existence of foliations of the n -torus by minimal hypersurfaces. We show that for a generic metric, whenever a lamination by area-minimizing hypersurfaces of the n -torus contains a gap, there exists a minimal hypersurface inside the gap that is not area-minimizing.

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1. INTRODUCTION AND MAIN RESULTS

In this paper, we study minimal hypersurfaces in the n -torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ equipped with a metric g . This problem may be viewed as a parametric version of the Aubry-Mather theory, in which area-minimizing hypersurfaces play a

role analogous to that of minimal orbits in convex Hamiltonian systems. In two dimensions, most of the results on Hamiltonian twist maps carry over to the case of minimizing geodesics on a two-torus (cf. [8]).

In 1986, Moser [36] first generalized the Aubry-Mather theory to higher dimensions. He studied minimizing graphs $u : \mathbb{R}^n \rightarrow \mathbb{R}$ of a $\mathbb{Z}^n$ -periodic variational integral. This may be regarded as a PDE analogue of the Aubry-Mather theory. A related problem, concerning minimizers of the elliptic integral on Caccioppoli sets, was studied by Caffarelli and de la Llave in [12]. In both settings, the authors established the existence of *plane-like* minimizers, namely minimizers contained in a strip of uniform bounded width between two parallel affine planes.

On a torus with an arbitrary metric, however, area-minimizing hypersurfaces need not be graphs. Auer and Bangert [5] and Auer [6] observed that tools from geometric measure theory could provide a natural framework in this setting. Roughly speaking, after passing to the universal cover, each area-minimizing hypersurface with the Birkhoff property (see Section 4) admits a unique homological direction $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$. Let $\mathcal{M}_\alpha$ denote the family of such hypersurfaces with direction α . They proved that hypersurfaces in $\mathcal{M}_\alpha$ with the same asymptotic behavior form a lamination on $\mathbb{R}^n$ whose projection is a minimal lamination on the torus (see Sections 4 and 5). A fundamental and important question, however, is whether such a lamination is a foliation. This happens in some exceptional cases, such as flat tori, but fails in general for certain metrics constructed by Bangert in [11].

A main goal of this paper is to use the Almgren-Pitts min-max theory to derive criteria for the existence of minimal foliations by hypersurfaces in $\mathcal{M}_\alpha$. For an integral class α , a standard Lusternik-Schnirelmann theory-type argument (see Appendix B) shows that $\mathbb{T}^n$ is foliated by a one-parameter family of closed minimal hypersurfaces in the class α exactly when

$$\omega(\alpha) - S(\alpha) = 0,$$

where $\omega(\alpha)$ is the min-max width of the class α and $S(\alpha)$ is its stable norm. We refer readers to Sections 2 and 4 below for the corresponding definitions.

We aim to extend this picture to irrational homology classes, where the width is not defined. To overcome this, we study the asymptotic behavior of the quantity $\omega(r) - S(r)$ as the integer class r approaches α .

In the case of geodesics on a two-dimensional torus, such a criterion was obtained by Mather in [31]: given a rational direction $r = [p, q] \in H_1(\mathbb{T}^2, \mathbb{Z})$ with p, q relatively prime, let ΔW_r be the infimum of all $\epsilon > 0$ such that any two periodic geodesics $\gamma_0, \gamma_1 \in \mathcal{M}_r$ can be joined by a continuous family γ_t of (p, q) -periodic curves satisfying

$$L(\pi(\gamma_t)) \leq L(\pi(\gamma_0)) + \epsilon \quad \text{for all } t,$$

where $\pi : \mathbb{R}^2 \rightarrow \mathbb{T}^2$ is the projection. Mather proved that for every irrational direction $\alpha \in H_1(\mathbb{T}^2, \mathbb{R})$, $\mathcal{M}_\alpha$ is a foliation if and only if:

$$\Delta W_\alpha := \lim_{p/q \rightarrow \alpha} \Delta W_r = 0.$$

Mather's proof was analytic and was formulated in the context of twist maps on a torus, which has no immediate generalization to higher dimensions. Our first main theorem utilizes the tools from geometric measure theory and min-max theory to generalize it to the case of hypersurfaces on the higher-dimensional n -torus, with $3 \leq n \leq 7$. The dimensional restriction comes from the regularity theory for minimal hypersurfaces.

Definition 1.1. Throughout the paper, we say that a sequence $\{r_k\}_k$ of primitive integer homology classes converges to a class $\alpha \neq 0$ if the normalized sequence $\frac{r_k}{S(r_k)}$ converges to $\frac{\alpha}{S(\alpha)}$ on the unit ball of the stable norm.

In light of Mather's criterion, we define the following:

Definition 1.2. For an integer class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, we define its *Mather-type energy barrier* as

$$\Delta W_r := \omega(r) - S(r).$$

We extend this energy to irrational classes as follows:

$$\Delta W_\alpha := \liminf_{\substack{r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z}) \\ r \rightarrow \alpha}} (\omega(r) - S(r)).$$

Here, the finiteness of the right-hand side follows from Proposition 2.5.

This should be viewed as a higher-dimensional analogue of the ϵ above. As discussed, this energy conceptually measures the obstruction to moving through the class r by almost area-minimizing hypersurfaces. Our first main result shows that, for totally irrational classes, the vanishing of this gap asymptotically characterizes minimal foliations.

Theorem 1.3. Suppose that $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ is totally irrational (see Definition 4.8). The following statements are equivalent.

- (1) $\lim_{k \rightarrow \infty} \Delta W_{r_k} = 0$ for all sequences of integer classes $\{r_k\}_k$ converging to α .
- (2) $\lim_{k \rightarrow \infty} \Delta W_{r_k} = 0$ for some sequence of integer classes $\{r_k\}_k$ converging to α .
- (3) The elements of $\mathcal{M}_\alpha$ form a foliation of $\mathbb{R}^n$ by area-minimizing hypersurfaces. Moreover, this foliation is invariant under the deck group and descends to a minimal foliation of $\mathbb{T}^n$.

The following is a direct consequence of Theorem 1.3.

Corollary 1.4. For $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ that is totally irrational, the torus is foliated by area-minimizing hypersurfaces in $\mathcal{M}_\alpha$ if and only if $\Delta W_\alpha = 0$.

We also mention the work of Aubry and Le Daeron [3] in connection with the Hamiltonian twist maps. Motivated by a problem in solid-state physics, they introduced a quantity called the *Peierls's barrier* and obtained a conclusion similar to that of Mather and ourselves. They showed that, for a given frequency, this quantity vanishes if and only if there is an invariant circle. Indeed, the Peierls's barrier is a lower bound of the Mather energy barrier defined above.

We highlight several points in Theorem 1.3. First, the implication (1) $\implies$ (2) is immediate. Second, it will follow from the proof in Section 6 that the implication (2) $\implies$ (3) holds for any homology class $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$. Finally, the implication (3) $\implies$ (1) does not hold in general if α is rationally dependent, as demonstrated in the following theorem.

Theorem 1.5. *For any nonzero homology class $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ such that $\alpha^\perp \cap (\mathbb{Z}^n \setminus \{0\}) \neq \emptyset$, there exists a Riemannian metric g on $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ and a sequence $\{r_k\}_k$ of integer classes converging to α such that*

$$\liminf_{r_k \rightarrow \alpha} \Delta W_{r_k} > 0,$$

yet $\mathbb{R}^n$ admits a foliation by area-minimizing hypersurfaces in $\mathcal{M}_\alpha$.

Nonetheless, for such α , the convergence of ΔW_{r_k} to 0 does hold under a stronger condition that the foliation in $\mathcal{M}_\alpha$ is unique. Let $\mathcal{M}(\alpha) \subset \mathcal{M}_\alpha$ be the subset consisting of the elements with maximal periodicity (see Section 4).

Theorem 1.6. *For any homology class $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$, if $\mathcal{M}(\alpha)$ is a foliation, then for any sequence of integer classes $\{r_k\}_k$ converging to α , we have $\lim_{k \rightarrow \infty} \Delta W_{r_k} = 0$.*

For integral classes α , the necessity of $\mathcal{M}(\alpha)$ being a foliation is immediate: one may simply take the constant sequence $r_k = \alpha$ for all k . We emphasize, however, that the converse is false. Indeed, one can construct a metric on $\mathbb{T}^n$ with the following property: for every primitive homology class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, except one, there exists a foliation of $\mathbb{T}^n$ by closed, homologically area-minimizing hypersurfaces in r . See Example 7.6.

In the case where the minimizing hypersurfaces do not fill all of $\mathbb{R}^n$ (that is, they form a genuine lamination with gaps), we prove that inside every gap, there is a complete minimal hypersurface that is not area-minimizing. This is an analogue of the following fact from the Aubry-Mather theory for the double-well potential shown in [23]: if the set of minimizing solutions (also called the ground states) does not attain all possible values at a point, then there is another non-minimizing critical point in between. In the case of variational monotone lattice recurrent relations, Mramor and Rink [37] proved a similar result: when an Aubry-Mather set has a gap, then this gap must be foliated by minimizers or contain a non-minimizing critical point. These critical points are often viewed as secondary laminations inside the gap.

Recall that a Riemannian metric g on a closed manifold M is called *bumpy* if every immersed, closed minimal hypersurface is non-degenerate (having no non-trivial Jacobi field). Brian White proved in [48] that bumpy metrics are generic in the sense of the Baire category.

Theorem 1.7. *Suppose that the metric g on $\mathbb{T}^n$ is bumpy and α is a totally irrational class. Assume further that elements of $\mathcal{M}_\alpha$ do not foliate $\mathbb{R}^n$. Then for any two consecutive minimizers Σ_0 and Σ_1 in $\mathcal{M}_\alpha$ that bound a gap G , there exists a complete, non-compact, embedded minimal hypersurface lying entirely inside G that is not area-minimizing and has index at most one.*

For geodesics on a two-torus, the existence of a similar non-area-minimizing critical point inside every gap was proved by Mather in [31]. His construction, in a more general setting of a monotone twist map on tori, used a minimax principle formulated by Birkhoff. Mather interpreted this critical point as a secondary orbit that is *homoclinic* to the original Cantor set (the lamination formed by recurrent minimizers). The minimal hypersurface constructed in Theorem 1.7 is a higher-dimensional analogue of this secondary orbit.

Nevertheless, our approach to constructing a mountain-pass type critical point inside the gap is different from Mather's. Roughly speaking, we construct such hypersurfaces as limits of sequences of index-one minimal hypersurfaces with prescribed boundaries on larger and larger compact exhaustions. This strategy is similar to the one employed by Carlotto and De Lellis [13] in their construction of min-max geodesics on asymptotically conical surfaces.

Although the minimal hypersurface constructed in Theorem 1.7 is non-compact, we can still describe its area asymptotically inside a large compact ball, in comparison to that of an area-minimizer. Let $B_R = B_R(p)$ be the ball centered at a point p of radius R with respect to a metric $\tilde{d}$ (see Section 9 for the definition).

Theorem 1.8. *Under the assumptions of Theorem 1.7, suppose that Γ is a non-area-minimizing minimal hypersurface constructed inside the gap G . The following holds.*

- (1) *If there is no closed minimal hypersurface inside G , then*

$$\limsup_{R \rightarrow \infty} (\mathbf{M}(\Gamma \llcorner B_R(p)) - \mathbf{M}(\Sigma_0 \llcorner B_R(p))) \geq \Delta W_\alpha.$$

- (2) *Otherwise, there exists a disjoint collection of closed, connected, stable minimal hypersurfaces $\mathcal{S} = \{S_1, S_2, \dots, S_j\}$ inside G and pairwise disjoint open sets $U_{S_1}, U_{S_2}, \dots, U_{S_j}$ with $S_i = \partial U_{S_i}$ such that either*

$$\sum_{i=1}^j W_{\partial}(U_{S_i}) \geq \Delta W_\alpha$$

or

$$\limsup_{R \rightarrow \infty} (\mathbf{M}(\Gamma \lfloor B_R) - \mathbf{M}(\Sigma_0 \lfloor B_R)) \geq \Delta W_\alpha - \sum_{i=1}^j \mathbf{M}(S_i).$$

Here, $W_\partial(U_{S_i})$ denotes the width of the manifold U_{S_i} with the strictly stable boundary S_i , as defined in Section 2.

1.1. Examples and related works.

1.1.1. *Minimal lamination.* On a flat torus, every direction determines a minimal foliation by parallel hyperplanes. The question of whether such foliations exist for general metrics was introduced by Moser in [36] and Bangert in [10].

In [34], Moser was interested in studying the stability of such minimal foliations. He called a minimal foliation F_g *stable* if there exists a C^∞ -neighborhood U of the metric g such that for every $g' \in U$, there exists a foliation $F_{g'}$ that is minimal with respect to g' and diffeomorphic to F_g .

On a flat torus, when α is an integral class, the minimal foliation by hyperplanes $x \cdot \alpha = \beta, \beta \in \mathbb{R}$ is not stable: arbitrarily small perturbations can break it into only finitely many closed area-minimizing hypersurfaces in the same homology class. In fact, this finite behavior is generic for primitive integral classes.

On the other hand, Moser proved that for $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n) \in \mathbb{R}^n$ which satisfies the following Diophantine condition

$$\sum_{i,j=1}^n (\alpha_i d_j - \alpha_j d_i)^2 \geq \gamma \left(\sum_{j=1}^n d_j^2 \right)^{-\tau}, \forall d = (d_1, d_2, \dots, d_n) \in \mathbb{Z}^n \setminus \{0\}, \quad (1.1)$$

for some constants $\gamma, \tau > 0$, the minimal foliation by affine hyperplanes in the direction α is indeed stable. Geometrically, this condition means that all integer points stay uniformly away from the line spanned by α .

We also note that Bangert [11] constructed $\mathbb{Z}^n$ -periodic metrics for which no minimal foliation exists. For these metrics, it follows from Theorem 1.3 that there exists a constant $c > 0$ such that $\omega(r) - S(r) > c$ for all $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$.

A related inverse question is when a given foliation can be made minimal for some Riemannian metric, that is, when it is *taut*. Sullivan [47] proved that a codimension-one oriented foliation admits such a metric if and only if every compact leaf is cut by a closed transversal curve.

1.1.2. *Differentiability of the stable norm.* In [4] and [18], Auer, Bangert, and Junginger-Gestrich investigated the differentiability properties of the stable norm

$$S : H_{n-1}(M, \mathbb{R}) \rightarrow \mathbb{R}.$$

They showed that the restriction of S to the minimal subspace spanned by the integer classes and α (see Definition 4.8) is differentiable at α . In particular, it is always differentiable at totally irrational classes. For rationally dependent

classes, S is differentiable at α exactly when $\mathcal{M}(\alpha)$ is a minimal foliation. More precisely, if $\mathcal{M}(\alpha)$ is not a foliation, heteroclinic minimizers in different directions in $\mathcal{M}_\alpha$ give rise to different subderivatives of S at α .

It is a consequence of these works that we can detect minimal foliations in rationally dependent classes consisting of the most periodic minimizers by examining the differentiability of the stable norm. It remains unclear, however, what the corresponding picture is for totally irrational classes.

1.1.3. Unstable minimal hypersurfaces on torus. In a flat three-torus, index-one minimal surfaces have been extensively studied. Ross [40] showed that the classical Schwarz $\mathcal{P}$ and $\mathcal{D}$ minimal surfaces, as well as A. Schoen's gyroid surface, all have Morse index one on the flat torus given by the quotient of the flat $\mathbb{R}^3$ by the corresponding lattice groups. In [20], the authors constructed (possibly non-orientable) minimal surfaces with index less than or equal to one on any flat torus. On the other hand, Ritoré and Ros [39] showed that the moduli spaces of orientable index-one minimal surfaces with genus greater than one in (non-fixed) flat three-tori are compact. As a consequence, on flat tori with prescribed volume, orientable index-one minimal surfaces do not exist when the injectivity radius is sufficiently small.

All known index-one examples listed above are homologically trivial. Indeed, by a theorem of Meeks [49], any closed two-sided minimal surface that is not totally geodesic in a flat three-torus is a Heegaard surface, hence null homologous (the same statement holds for any closed manifold of nonnegative Ricci curvature). On the other hand, totally geodesic minimal surfaces on a flat three-torus are flat two-tori, and they foliate the torus for each non-trivial homology class.

One can also see the non-existence of a connected, unstable minimal surface in any non-trivial homology class by the maximum principle.

It is currently unknown if there exist infinitely many connected, embedded, two-sided, index-one minimal surfaces on the cubic flat three-torus.

1.2. Plan of the paper. In Section 2, we briefly review the necessary notation and the basic framework of one-parameter Almgren-Pitts min-max theory. We use that to define the *width* associated with an integer homology class. In Section 3, we reformulate an interpolation theorem due to Marques-Neves in a form that allows us to construct a mass-continuous map interpolating between two currents that are close in the mass norm. This is a crucial tool in the proofs of Theorems 1.3 and 1.6. The detailed proofs of these results are carried out in Section 6.

We briefly summarize the theory of area-minimizing hypersurfaces in tori in Section 4, with a continued focus on irrational homological direction in Section 5. In Section 7, we use a Lusternik-Schnirelmann-type argument to derive a sufficient condition for the existence of a foliation in a given homology class. Examples of metrics that satisfy Theorem 1.5 are constructed in Section 8.

Section 9 contains the proof of Theorem 1.7 and Theorem 1.8, in which we construct and analyze the asymptotic area of complete, non-compact minimal hypersurfaces that are not area-minimizing inside the gaps of an irrational lamination, using a version of min-max theory with boundary developed by Montezuma. We conclude with a discussion of some open problems in the last section.

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2. ALMGREN-PITTS MIN-MAX THEORY

2.1. The width of an integer homology class. In this section, we recall some standard definitions from geometric measure theory. Most of these can be found in the book of Simon [22]. Let M^n be a closed n -dimensional manifold equipped with a Riemannian metric g . We assume that (M^n, g) is isometrically embedded in some Euclidean space $\mathbb{R}^J$. For $0 \leq k \leq n$, denote by $\mathcal{I}_k(M)$ the space of k -dimensional flat chains with integer coefficients in M and let $\mathcal{Z}_{n-1}(M)$ be the space of all $(n-1)$ -dimensional integral cycles T in $\mathcal{I}_{n-1}(M)$ satisfying $T = \partial Q$ for some $Q \in \mathcal{I}_n(M)$. This is precisely the connected component of the space of integral currents without boundary that contains the zero current.

For any $T \in \mathcal{I}_k(M)$, we write $|T|$ and $\|T\|$ for the integral varifold and the Radon measure in M associated with T . The mass of a current T is defined as:

$$\mathbf{M}(T) = \sup\{T(\phi) : \phi \in \mathcal{D}^k(\mathbb{R}^J), \|\phi\| \leq 1\},$$

where $\mathcal{D}^k(\mathbb{R}^J)$ is the space of smooth k -forms with compact support.

The flat metric on $\mathcal{I}_k(M)$ is defined as:

$$\mathcal{F}(S, T) = \inf\{\mathbf{M}(P) + \mathbf{M}(Q) : S - T = P + \partial Q, P \in \mathcal{I}_k(M), Q \in \mathcal{I}_{k+1}(M)\}.$$

The $\mathbf{F}$ -metric on the closure $\mathcal{V}_k(M)$ of the space of k -dimensional rectifiable varifolds with support contained in M is defined by:

$$\mathbf{F}(V, W) = \sup\{V(f) - W(f) : f \in C_c(G_k(\mathbb{R}^J)), |f| \leq 1, \text{Lip}(f) \leq 1\}.$$

This induces the following $\mathbf{F}$ -metric on $\mathcal{I}_k(M)$:

$$\mathbf{F}(S, T) = \mathcal{F}(S, T) + \mathbf{F}(|S|, |T|).$$

A continuous map in the $\mathcal{F}$ -topology

$$\Phi : S^1 \rightarrow \mathcal{Z}_{n-1}(M), \quad \Phi(0) = \Phi(2\pi) = 0$$

is called a *sweep-out* if it has no concentration of mass and represents a nontrivial element in $\pi_1(\mathcal{Z}_{n-1}(M), \{0\})$. The first condition means the following:

$$\limsup_{r \rightarrow 0} \{ \|\Phi(x)\|(B_r(p)) : x \in S^1, p \in M \} = 0$$

and the second condition is equivalent to the image of Φ under the Almgren isomorphism

$$F : \pi_1(\mathcal{Z}_{n-1}(M), \{0\}) \rightarrow H_n(M, \mathbb{Z})$$

being the fundamental class $[M]$. Let $\mathcal{P}$ denote the collection of all such sweep-outs. The width of M is then defined as follows:

$$\omega(M) := \inf_{\mathcal{P}} \sup_{\theta \in S^1} \mathbf{M}(\Phi(\theta)). \quad (2.1)$$

It can be deduced from the isoperimetric inequality that the width $\omega(M)$ is always positive. Almgren [2] established the existence of a stationary integral varifold realizing $\omega(M)$, and Pitts [38] later proved that this varifold is actually realized by an embedded minimal hypersurface. Moreover, on a manifold equipped with a generic metric, this min-max minimal hypersurface is unstable and has Morse index one.

This setting can be applied in a similar manner to find an unstable index-one minimal hypersurface in any integer homology class. Let $\mathcal{Z}_{n-1}(M, r)$ denote the space of all integral cycles that represent the same homology class $r \in H_{n-1}(M, \mathbb{Z})$ in M . This is the connected component containing any fixed cycle $S \in r$ of the space of all cycles. Naturally, $\mathcal{Z}_{n-1}(M, r)$ is homeomorphic to $\mathcal{Z}_{n-1}(M)$ via the translation map sending T to $T - S$.

Let $\mathcal{P}_r$ be the collection of all sweep-outs of M by cycles in $\mathcal{Z}_{n-1}(M, r)$. The *min-max width of the class r* is defined as follows:

$$\omega(r) := \inf_{\mathcal{P}_r} \sup_{\theta \in S^1} \mathbf{M}(\Phi(\theta)).$$

If we want to emphasize the dependence on the Riemannian metric g , we shall write $\omega_g(r)$ instead.

It follows from the works of Marques-Neves [27] that the following holds:

Theorem 2.1. *Let (M^n, g) , with $3 \leq n \leq 7$, be a closed manifold and r be a fixed primitive homology class. Then there exists a stationary integral varifold V whose support is a closed embedded minimal hypersurface Σ of index bounded by one, such that*

$$\|V\|(M) = \omega(r).$$

Furthermore, we have the decomposition

$$\Sigma = n_1 \Sigma_1 + \cdots + n_k \Sigma_k$$

where each Σ_i is a closed, connected, and embedded minimal hypersurface and $n_i \in \mathbb{Z}_{>0}$. Any two Σ_i, Σ_j with $i \neq j$ are disjoint, and:

$$\text{Ind}(\Sigma) = \sum_{i=1}^k \text{Ind}(\Sigma_i) \leq 1.$$

By the definition of the stable norm, we have $\mathbf{M}(\Phi(\theta)) \geq S(r)$ for all $\theta \in S^1$ and $\Phi \in \mathcal{P}_r$. As a consequence

$$\omega(r) \geq S(r), \quad \forall r \in H_{n-1}(M, \mathbb{Z}).$$

A standard Lusternik-Schnirelmann theory-type argument shows that equality occurs if and only if M is foliated by area-minimizing hypersurfaces in the class r . See Appendix B for more details.

Because we are considering one-parameter families of hypersurfaces lying in the homology class r , the support of the min-max limit Σ necessarily contains at least one closed hypersurface in r . This is made precise in the following proposition:

Proposition 2.2. *In the setup of Theorem 2.1, the support of the min-max minimal hypersurface Σ contains a closed embedded minimal hypersurface (not necessarily connected) that represents the homology class r .*

Proof. The statement follows from examining the convergence of a min-max sequence in two different senses. Suppose that $\{|\Phi_i(x_i)|\}_i$ is a min-max sequence converging in the varifold sense to

$$|\Sigma| = \sum_{i=1}^l m_i |\Sigma_i|.$$

It follows from the continuity of mass with respect to the varifold convergence that $\mathbf{M}(\Phi_i(x_i))$ is uniformly bounded. Therefore, the weak compactness theorem for integral currents [22] implies that there is a subsequence (still denoted by $\Phi_i(x_i)$) that converges in the flat topology to some integral current T :

$$\lim_{i \rightarrow \infty} \Phi_i(x_i) = T.$$

Varifold convergence also implies that the associated Radon measure $||\Phi_i(x_i)||$ converges to $||\Sigma||$ weakly. Therefore, the support of T must be contained in the support of Σ , meaning

$$\text{spt}(T) \subset \text{spt}(\Sigma) = \bigcup_{i=1}^l \Sigma_i.$$

Since $\Phi_i(x_i)$ represents the same homology class r for every i , the same is true for T . The proposition follows. $\square$

Remark 2.3. A basic difficulty that arises when trying to produce an index-one minimal hypersurface in a prescribed homology class r by this method is the following. It may happen that the minimal hypersurface in r realizing the width is not connected and instead consists of several disjoint, connected components. In such a situation, one component may be stable and represent the homology class r , while another component may have index one but be homologically trivial.

To illustrate this phenomenon, consider a flat torus to which a large sphere is attached (see Figure 1 below). For any nontrivial class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, the minimal hypersurface that realizes $\omega(r)$ has two connected components: one is an area-minimizing hypersurface in r , and the other is an index-one great equator arising from the spherical part. Consequently, the one-parameter

family min-max construction does not yield a connected, index-one minimal surface in any integer class.

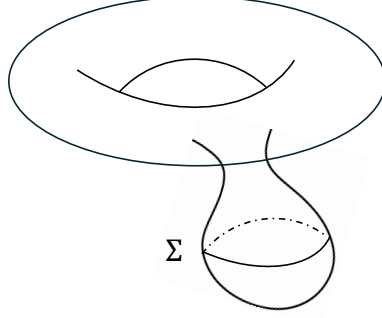


FIGURE 1. Torus with a sphere attached. The component Σ represents an index-one equator arising from the spherical region.

Nonetheless, when the quantity ΔW_r is sufficiently small, there is not enough excess area for an additional closed component. Consequently, the min-max minimal hypersurface in the class r must consist of a single connected component representing r .

Proposition 2.4. *Suppose that $3 \leq n \leq 7$ and that g is a Riemannian metric on $\mathbb{T}^n$. There exists a positive constant C , depending only on the metric g , with the following property: for every primitive homology class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ satisfying $\Delta W_r < C$, there exists a connected minimal hypersurface Σ_r representing the class r for which the min-max width $\omega(r)$ is achieved by Σ_r .*

Proof. By the monotonicity formula for minimal submanifolds, there exists a constant $C = C(g)$ such that for every closed minimal hypersurface Σ on $\mathbb{T}^n$, we have:

$$\mathbf{M}(\Sigma) > C.$$

We will show that this constant satisfies the proposition. Suppose that $\Delta W_r = \omega(r) - S(r) < C$. By Theorem 2.1, there exist connected, pairwise disjoint minimal hypersurfaces $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ and positive integers $m_1, m_2, \dots, m_k$ such that

$$\omega(r) = m_1 \mathbf{M}(\Sigma_1) + \dots + m_k \mathbf{M}(\Sigma_k).$$

Since $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ is primitive, it follows from Proposition 2.2 and the fact that any two hypersurfaces representing independent homology classes intersect (see Appendix A), at least one of the $\Sigma_i, 1 \leq i \leq k$ must belong to the class r . Without loss of generality, assume that $\Sigma_1 \in r$. From the definition of

the stable norm, we have that $S(r) \leq \mathbf{M}(\Sigma_1)$. As a result:

$$\begin{aligned} \Delta W_r = \omega(r) - S(r) &= (\mathbf{M}(\Sigma_1) - S(r)) + (m_1 - 1)\mathbf{M}(\Sigma_1) + \sum_{i=2}^k \mathbf{M}(\Sigma_i) \\ &\geq (m_1 - 1)\mathbf{M}(\Sigma_1) + \sum_{i=2}^k \mathbf{M}(\Sigma_i). \end{aligned} \quad (2.2)$$

By the choice of C , we have that $\mathbf{M}(\Sigma_i) > C$ for all $i = 1, 2, \dots, k$. Together with the inequality $\Delta W_r < C$, it must happen that $k = m_1 = 1$ and we conclude that $\omega(r) = \mathbf{M}(\Sigma_1)$. The proof is complete. $\square$

It is therefore of interest to bound the quantity ΔW_r from above. Recall that $\omega(M)$ denotes the usual width of M (or, in this context, the width of the trivial class). We have the following upper bound that holds for any Riemannian metric on any Riemannian manifold of any dimension.

Proposition 2.5. *For any integer homology class $r \in H_{n-1}(M, \mathbb{Z})$, we have:*

$$0 \leq \omega(r) - S(r) \leq \omega(M).$$

Proof. For a given $\epsilon > 0$, we can choose a sweep-out $\Phi : S^1 \rightarrow \mathcal{Z}_{n-1}(M)$ such that

$$\sup_{t \in S^1} \mathbf{M}(\Phi(t)) \leq \omega(M) + \epsilon.$$

Consider the map ϕ defined as

$$\phi(t) = \Phi(t) + T_r,$$

where $T_r \in r$ is a fixed mass-minimizing integer current in r . Since $[\Phi(t)] = 0 \in H_{n-1}(M, \mathbb{Z})$, we have $\phi(t) \in r$. It follows that ϕ is a sweep-out of M by currents in $\mathcal{Z}_{n-1}(M, r)$. Note that

$$\begin{aligned} \mathbf{M}(\phi(t)) &= \mathbf{M}(\Phi(t) + T_r) \\ &\leq \mathbf{M}(\Phi(t)) + \mathbf{M}(T_r) \\ &\leq \sup \mathbf{M}(\Phi(t)) + S(r) \\ &\leq \omega(M) + \epsilon + S(r) \end{aligned} \quad (2.3)$$

for all $t \in [0, 1]$. Therefore:

$$\omega(r) \leq \sup \mathbf{M}(\phi(t)) \leq \omega(M) + \epsilon + S(r).$$

Sending ϵ to 0, we establish the desired inequality. $\square$

A similar argument yields the following more general inequality:

$$\omega((k+1)r) - S((k+1)r) \leq \omega(kr) - S(kr)$$

where $kr \in H_{n-1}(M, \mathbb{Z})$ is an integer multiple of the class r .

It follows that for any sequence $\{r_k\}_k$ of integer classes, the sequence

$$\Delta W_{r_k} = \omega(r_k) - S(r_k)$$

is bounded and thus has a convergent subsequence. Theorem 1.3 states that if α is totally irrational and $\mathcal{M}_\alpha$ is a foliation, then for any sequence of integer classes $r_k \rightarrow \alpha$, we have $\Delta W_{r_k} \rightarrow 0$. Conversely, if there exists one sequence $r_k \rightarrow \alpha$ for which $\Delta W_{r_k} \rightarrow 0$, then the same conclusion holds for every such sequence.

We expect that, more generally, the limit of ΔW_{r_k} as $r_k \rightarrow \alpha$ exists and is independent of the particular sequence, provided that α is totally irrational. In other words, we hope to replace the limit inferior in the definition of the Mather energy barrier with the regular limit. In dimension two, the analogous result for $\mathbb{T}^2$ with geodesics was proved by Mather in [31].

2.2. A local min-max construction on manifolds with boundary. We give a brief overview of a local min-max construction that is discussed in [45], Appendix B. It is essentially contained in [27]. Let N^n be a compact, connected, orientable Riemannian manifold. Suppose that

$$\partial N = \partial_0 N \cup \partial_1 N$$

where $\partial_0 N$ and $\partial_1 N$ (which can be empty) are two disjoint, closed components.

We are interested in sweep-outs of N that connect $\partial_0 N$ to $\partial_1 N$:

$$\Phi : [0, 1] \rightarrow \mathcal{Z}_{n-1}(N; \mathbf{F}, \mathbb{Z}), \quad \Phi(0) = \partial_0 N, \quad \Phi(1) = \partial_1 N.$$

Let Π be the collection of all continuous sweep-outs Φ' that are homotopic to Φ in the flat topology, with fixed endpoints $\partial_0 N$ and $\partial_1 N$. The *width* of Π is defined as follows:

$$L(\Pi) = \inf_{\Phi \in \Pi} \sup_{t \in [0, 1]} \mathbf{M}(\Phi(t)).$$

The width of (N, g) is then defined as the infimum of $L(\Pi)$ over all possible homotopy classes of sweep-outs as above.

$$W_\partial(N) = \inf_{\Pi} L(\Pi).$$

One should not confuse this width with the one-width $\omega_1(N, g)$ defined using relative cycles.

The following min-max theorem is essentially contained in the work of Marques-Neves [27]. See also [45], Appendix B, for a proof.

Theorem 2.6. *Let $3 \leq n \leq 7$ and N^n be a compact, connected, orientable manifold endowed with a bumpy metric g . Assume that ∂N is a strictly stable minimal hypersurface. Then there exists a stationary integral varifold V whose support is a smooth embedded minimal hypersurface $\Sigma \in N$ of index at most one, such that*

$$||V||(N) = W_\partial(N).$$

Furthermore, at least one of the components of Σ is contained in the interior of N .

2.3. A mountain pass theorem for minimal hypersurfaces with prescribed boundary. We conclude this section by recalling the following version of min-max theory for minimal hypersurfaces with a prescribed boundary, developed by Montezuma in [33]. Roughly speaking, he showed that, under suitable additional hypotheses, between two strictly stable minimal hypersurfaces spanning the same boundary, there is a third minimal hypersurface of the mountain-pass type with the same boundary. This construction will play a central role in the proof of Theorem 1.7.

We now describe in detail the geometric setting in which the result applies. For $3 \leq n \leq 7$, let M^n be a compact Riemannian manifold whose boundary ∂M is strictly convex. Let $\gamma^{n-2} \subset \partial M$ be a smooth, oriented, closed $(n-2)$ -dimensional submanifold. Assume further that there exist two distinct, smooth, oriented, embedded minimal hypersurfaces Σ_0 and Σ_1 satisfying

$$\partial \Sigma_0 = \partial \Sigma_1 = \gamma,$$

and suppose that both are strictly stable and homologous.

In this version of min-max theory, one considers paths in the space $\mathcal{Z}_{n-1}(M, \gamma)$ of codimension-one integral currents in M with boundary supported in γ . For a homotopy class Π of such paths in $\mathcal{Z}(M, \gamma)$ joining Σ_0 and Σ_1 , the min-max invariant is defined by

$$\mathbf{L}(\Pi) := \inf_{\{\Sigma_t\} \in \Pi} \sup_{t \in [0,1]} \mathbf{M}(\Sigma_t).$$

Taking the infimum over all possible homotopy classes Π , one can define the following width:

$$W_\gamma(\Sigma_0, \Sigma_1) = \inf_{\Pi} \mathbf{L}(\Pi).$$

The following theorem is essentially a combination of Theorem 1.3 in [33] and a prescribed boundary version of Sharp's compactness theorem for minimal hypersurfaces with bounded Morse index (cf. [24], Theorem 4.1). In fact, a standard modification shows that one only needs to assume that $H_{\Sigma_0}, H_{\Sigma_1} \geq 0$ with respect to the inward-pointing normal vectors (cf. [29]).

Theorem 2.7. *Let M^n and γ^{n-2} be as above. Assume that*

$$W_\gamma(\Sigma_0, \Sigma_1) > \max\{\mathbf{M}(\Sigma_0), \mathbf{M}(\Sigma_1)\}.$$

Then there exists an integral varifold $V \in \mathcal{V}(M)$ whose support is the closure of an embedded smooth minimal hypersurface Γ with $\dim(\text{Sing}(\Gamma)) \leq n-8$. If $\Gamma_1, \Gamma_2, \dots, \Gamma_k$ denote the connected components of Γ , then:

- (1) $\text{Sing}(\Gamma_i) \cap M = \emptyset$, for every $i = 1, 2, \dots, k$;
- (2) each $\partial \Gamma_i$ is either the empty set or a union of components of γ . Moreover $\bigcup_{i=1}^k \partial \Gamma_i = \gamma$;
- (3) there are positive integers $m_1, m_2, \dots, m_k$, such that $V = \sum_{i=1}^k m_i \overline{\Gamma_i}$. The multiplicity of the component Γ_i with a non-empty boundary $\partial \Gamma_i$ is $m_i = 1$;

$$(4) \quad W_\gamma(\Sigma_0, \Sigma_1) = \|V\|(M) = \mathbf{M}(\Gamma) = \sum_{i=1}^k m_i \mathbf{M}(\Gamma_i).$$

3. AN INTERPOLATION LEMMA

In this section, we recall and slightly reformulate an interpolation theorem due to Marques-Neves in [26]. Its primary purpose is to approximate a discrete map of sufficiently small fineness (a notion defined below) by a continuous map in the mass norm. This is especially useful when one wishes to construct a one-parameter family of currents with controlled mass connecting two given currents T_1, T_2 . In particular, if T_1 and T_2 are mass-minimizing currents that are close in the flat topology, we can interpolate from one to the other continuously by a family of currents of similar masses.

This result will be applied in the proof of the implication (3) $\implies$ (1) in Theorem 1.3, where we need to construct near-optimal sweep-outs by integral currents in the class r_k to estimate the width $\omega(r_k)$.

We begin by recalling some additional notation that was introduced in [26]. For integers $m, j \geq 1$, let $I(m, j)$ be the cell complex on the unit m -dimensional cube I^m :

$$I(m, j) = I(1, j) \otimes I(1, j) \otimes \cdots \otimes I(1, j) \quad (\text{with } m \text{ factors}).$$

Here $I(1, j)$ is the cube complex on $I^1 = [0, 1]$ whose 1-cells and 0-cells are given by the subdivisions

$$[0, 3^{-j}], \dots, [1 - 3^{-j}, 1] \quad \text{and} \quad [0], [3^{-j}], \dots, [1].$$

The cube complex $X(j)$ is the union of all the cells of $I(m, j)$ whose support is contained in some cell of X , and $X(j)_0$ is the set of all 0-cells (vertices) in $X(j)$.

Given a discrete map $\phi : X(j)_0 \rightarrow \mathcal{Z}_{n-1}(M)$, we define the *fineness* of ϕ to be

$$\mathbf{f}(\phi) = \sup\{\mathbf{M}(\phi(x) - \phi(y)) : x, y \text{ adjacent vertices in } X(j)_0\}.$$

Theorem 3.1. *There exist positive constants C_0 and δ_0 depending only on (M^n, g) and m such that for every map*

$$\psi : I(m, j)_0 \rightarrow \mathcal{Z}_{n-1}(M)$$

with $\mathbf{f}(\psi) < \delta_0$, we can find a continuous map in the mass norm

$$\Psi : I^m \rightarrow \mathcal{Z}_{n-1}(M; \mathbf{M})$$

such that

- (1) $\Psi(x) = \psi(x)$ for all $x \in I(m, j)_0$;
- (2) for every $\alpha \in I(m, j)_p$, we have the following:

$$\sup\{\mathbf{M}(\Psi(x) - \Psi(y)) : x, y \in \alpha\} \leq C_0 \mathbf{f}(\psi).$$

Specializing this to the unit interval $I = [0, 1]$, we have the following:

Corollary 3.2. *For every $0 < \delta < \delta_0$ and for any $T_0, T_1 \in \mathcal{Z}_{n-1}(M)$ with $\mathbf{M}(T_0 - T_1) < \delta$, we can find a continuous map in the mass norm*

$$\Psi : [0, 1] \rightarrow \mathcal{Z}_{n-1}(M, \mathbf{M})$$

such that

- (1) $\Psi(0) = T_0, \Psi(1) = T_1$;
- (2) for all $t \in [0, 1]$ we have

$$\mathbf{M}(\Psi(t)) \leq \mathbf{M}(T_0) + C_0 \delta.$$

Proof. Consider the cell complex $I(1, 0)$ on I^1 with two vertices $[0]$ and $[1]$. Let $\psi(0) = T_0$ and $\psi(1) = T_1$. From the definition of fineness, we have

$$\mathbf{f}(\psi) = \mathbf{M}(T_1 - T_0) < \delta_0.$$

It follows from Theorem 3.1 that we can extend ψ to a mass-continuous map

$$\Psi : [0, 1] \rightarrow \mathcal{Z}_{n-1}(M; \mathbf{M})$$

so that

$$|\mathbf{M}(\Psi(t)) - \mathbf{M}(\Psi(0))| \leq |\mathbf{M}(\Psi(t) - \Psi(0))| \leq C_0 \delta$$

for all $t \in [0, 1]$. Since $\Psi(0) = T_0$, we are done. $\square$

4. AUBRY-MATHER THEORY FOR AREA-MINIMIZING HYPERSURFACES ON TORI

We now describe the setting in detail. Consider the n -dimensional torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ equipped with a Riemannian metric g and fix a Riemannian universal covering $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$. The metric g lifts to a $\mathbb{Z}^n$ -periodic metric (still denoted by g) on $\mathbb{R}^n$. Under the Hurewicz homomorphism, we have the identification $\pi_1(\mathbb{T}^n) \cong H_1(\mathbb{T}^n, \mathbb{Z})$, which acts as the group of deck transformations of π . The deck transformation associated with the element $k \in \pi_1(\mathbb{T}^n)$ is denoted by τ_k and we have $\tau_k(x) = x + k$ for all $x \in \mathbb{R}^n$.

A hypersurface S on $\mathbb{R}^n$ is said to have the *Birkhoff property* or is called *self-intersection free* if its projection to $\mathbb{T}^n$ has no transverse self-intersection. This is equivalent to the fact that for any integer translation τ_k of $\mathbb{R}^n$, either S is τ_k -invariant or S and $\tau_k(S)$ are disjoint.

We are interested in understanding the structure of the following set:

$$\mathcal{M} := \left\{ S = \partial \llbracket E \rrbracket \mid \begin{array}{l} S \text{ is self-intersection free} \\ \text{and area-minimizing and has connected support} \end{array} \right\}.$$

Here $E \subset \mathbb{R}^n$ is an open set of locally finite perimeter and S is area-minimizing if for all bounded open sets $U \subset \mathbb{R}^n$ we have

$$\mathbf{M}_U(S) \leq \mathbf{M}_U(S + T)$$

for all $T \in \mathcal{R}_{n-1}(U)$ with $\text{spt}(T) \subset U$ and $\partial T = 0$. Here, we always choose E consistently with the orientation of S .

There is the following natural partial order on $\mathcal{M}$:

$$\Sigma_0 = \partial E_0 < \Sigma_1 = \partial E_1 \iff E_0 \subset E_1.$$

For any $S = \partial[E] \in \mathcal{M}$, it was proved by Moser in [36] in the non-parametric case and Auer in [6] in the parametric analogue that there exists a homological direction $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R}) \cong \mathbb{R}^n$, unique up to scalar multiplication, such that the following holds:

$$\tau_k(E) \subset E \text{ if } k \cdot \alpha > 0, \quad (4.1)$$

$$E \subset \tau_k(E) \text{ if } k \cdot \alpha < 0 \quad (4.2)$$

for any $k \in \pi_1(\mathbb{T}^n) = \mathbb{Z}^n$. Let $\mathcal{M}_\alpha$ be the subset of $\mathcal{M}$ consisting of elements having direction α . Their results showed even more: an area-minimizing hypersurface in $\mathcal{M}_\alpha$ always stays within a universally bounded distance from an affine hyperplane.

Since the metric is periodic, there are positive constants λ and Λ depending only on g such that

$$\lambda g_{\text{flat}} \leq g \leq \Lambda g_{\text{flat}}.$$

Theorem 4.1. *There exists a constant $C > 0$ depending only on λ and Λ such that for every area-minimizing hypersurface $S = \partial E \in \mathcal{M}_\alpha$, there exists a positive constant $K = K(S)$ such that*

$$\text{spt}(S) \subset \left\{ x \in \mathbb{R}^n : x \cdot \frac{\alpha}{S(\alpha)} \in [K, K + C] \right\}.$$

We now briefly discuss the existence of area-minimizing hypersurfaces in each $\mathcal{M}_\alpha$. On manifolds with non-trivial homology groups, one can produce minimal submanifolds by minimizing the area functional within a fixed homology class. The existence and regularity for such minimizers were established by Federer in [21].

Theorem 4.2. *Let (M^n, g) be a closed Riemannian manifold, and let $\alpha \in H_k(M, \mathbb{Z})$ be a non-trivial homology class. Then there exists a closed minimal submanifold Σ smooth away from a singular set of dimension at most $k - 7$ that minimizes the area among all representatives of α . Moreover, the support of Σ consists of disjoint embedded minimal submanifolds, possibly occurring with multiplicities.*

In this paper, unless otherwise stated, we always work under the classical assumption $3 \leq n \leq 7$, so that area-minimizing hypersurfaces are smooth. Under the equivalence between singular homology and the homology of a chain complex of normal currents, see Theorem 5.11 of [16], Federer also proved that an area-minimizing hypersurface Σ minimizes the mass among all closed, normal currents representing α .

Definition 4.3. Following [21], the *stable norm* $S(\alpha)$ of a real homology class $\alpha \in H_m(M, \mathbb{R})$ is defined as the infimum of the volume of all real Lipschitz cycles

$$c = \sum r_i \sigma_i, \quad r_i \in \mathbb{R}, \quad \sigma_i : \Delta_i \rightarrow M$$

representing α . It was shown in [4] that

$$S(\alpha) = \min\{\mathbf{M}(T) \mid T \in \mathbf{N}_m(M), \partial T = 0, [T] = \alpha\},$$

where $\mathbf{N}_m(M)$ is the space of all normal currents in M . Recall that a current T is normal if both T and ∂T have finite mass.

In the context of the n -torus, we have that $H_{n-1}(\mathbb{T}^n, \mathbb{Z}) = \mathbb{Z}^n$. Furthermore, it was proved by Bangert and Auer in [5] that a homologically area-minimizing hypersurface Σ in a primitive class (defined as not being an integer multiple of another integer class) is connected. Specifically, they proved that the stable norm on a torus is always strictly convex:

$$S(\alpha + \beta) < S(\alpha) + S(\beta) \tag{4.3}$$

for all $\alpha, \beta \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ that are linearly independent. If a homologically area-minimizing hypersurface Σ in a primitive class γ decomposes as $\Sigma_1 + \Sigma_2$, then Σ_1 and Σ_2 minimize area in their respective homology classes. Let $[\Sigma_1] = \alpha$ and $[\Sigma_2] = \beta$. We have:

$$S(\alpha + \beta) = \mathbf{M}(\Sigma_1 + \Sigma_2) = \mathbf{M}(\Sigma_1) + \mathbf{M}(\Sigma_2) = S(\alpha) + S(\beta),$$

contradicting (4.3). A more direct argument showing the connectedness of Σ is given in Appendix A.

Therefore, for each $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, we can lift Σ to the universal cover $\mathbb{R}^n$ to obtain a connected hypersurface $\tilde{\Sigma}$ in the direction α :

Definition 4.4. Assume that $p : \widehat{M}^n \rightarrow M^n$ is a normal Riemannian covering and T is an m -current of locally finite mass on M^n . We can define the lift $p^\#T$ of T as a current of locally finite mass on the cover as follows: for every open set $U \subset \widehat{M}^n$ such that $p|_U$ is injective and for every compactly supported m -form ω on U , set

$$p^\#T(\omega) = ((p|_U)^{-1})_\#(T)(\omega),$$

where $(p|_U)^{-1}$ denotes the local inverse of p , and $\#$ is the standard push-forward operation of a current. We can extend this definition by a partition of unity.

It follows immediately from the definition that $\tilde{\Sigma}$ is area-minimizing on every compact set that projects injectively to the torus. In fact, using a result of Federer [21], one can show that $\tilde{\Sigma}$ is area-minimizing on every compact subset of $\mathbb{R}^n$:

Proposition 4.5. *Let $U \subset \mathbb{R}^n$ be a bounded open set and S be a current with support contained inside U and $\partial S = 0$. Then*

$$\mathbf{M}_U(\tilde{\Sigma}) \leq \mathbf{M}_U(\tilde{\Sigma} + S).$$

Proof. Since $H_{n-1}(\mathbb{R}^n, \mathbb{Z}) = 0$, there exists W with support contained inside U and $S = \partial W$.

Recall that $\tilde{\Sigma}$ is a lift of a homologically area-minimizing hypersurface Σ on $\mathbb{T}^n$. Let $H \subset \mathbb{R}^n$ be a fundamental domain. Since we can choose a sufficiently large integer k such that U is contained inside the union of k isometric copies of H , we can assume that U is such a union. It follows from a standard result of Federer [21] that $k\Sigma$ is homologically area-minimizing in the class $k[\Sigma]$. Therefore

$$\mathbf{M}_U(\tilde{\Sigma}) = \mathbf{M}(k\Sigma) \leq \mathbf{M}(k\Sigma + \partial\pi(W)) = \mathbf{M}_U(\tilde{\Sigma} + \partial W) = \mathbf{M}_U(\tilde{\Sigma} + S).$$

This completes the proof. $\square$

The remaining fact that the homological direction of $\tilde{\Sigma}$ is α up to rescaling follows from the following claim:

Lemma 4.6. ([4], Lemma 5.4). *Suppose that $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ is a real homology class, and let $T \in \mathbf{N}_{n-1}(\mathbb{T}^n)$ be a closed normal current representing α . Let $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$ be the universal covering map realizing $H_{n-1}(\mathbb{T}^n, \mathbb{R}) \cong \mathbb{R}^n$. Then there exists a function f of locally bounded variation on $\mathbb{R}^n$ such that the following hold:*

- (1) *for any $(n-1)$ -form ω on $\mathbb{R}^n$ with compact support, we have:*

$$T(\omega) = \int_{\mathbb{R}^n} f d\omega,$$

- (2) *for all $x \in \mathbb{R}^n$ and $k \in \pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$, we have:*

$$f(\tau_k x) = f(x) - k \cdot \alpha.$$

In particular, every leaf $\tilde{T}$ of $\pi^\# T$ has the Birkhoff property and satisfies

$$\tilde{T} < \tau_k \tilde{T} \quad \text{if and only if} \quad k \cdot \alpha > 0,$$

$$\tilde{T} = \tau_k \tilde{T} \quad \text{if and only if} \quad k \cdot \alpha = 0.$$

This shows that $\tilde{\Sigma} \in \mathcal{M}_\alpha$. In fact, it belongs to the subset consisting of the most periodic elements in the hierarchy described below.

On the other hand, when the class α is irrational (see the definition below), we can take a sequence of integer classes $\{\alpha_i\}_i$ converging to α and a sequence of corresponding periodic minimizers $\Sigma_i \in \mathcal{M}(\alpha_i)$ that intersect a fixed compact set (this is possible by considering the integer translations of a fixed minimizer). The following standard compactness theorem for area-minimizing hypersurfaces shows that we can take the limit of the sequence $\{\Sigma_i\}_i$ to obtain an element in $\mathcal{M}_\alpha$:

Theorem 4.7. *For $3 \leq n \leq 7$, the set of minimizing hypersurfaces $\mathcal{M}$ is compact with respect to the C^k convergence on compact sets, for every k . Equivalently, suppose that $\Sigma_i \in \mathcal{M}_{\alpha_i}$ is a sequence of such hypersurfaces, all of which intersect a fixed bounded set $K \subset \mathbb{R}^n$. Then there exists a subsequence (still denoted by $\{\Sigma_i\}_i$) that converges to a minimizing hypersurface $\Sigma \in \mathcal{M}$.*

Furthermore, the homological vectors respect this convergence: $\Sigma \in \mathcal{M}_\alpha$ where α is the limit of the sequence $\{\alpha_i\}_i$.

In particular, $\mathcal{M}_\alpha \neq \emptyset$ for all α and we have the following decomposition:

$$\mathcal{M} = \bigsqcup_{S(\alpha)=1} \mathcal{M}_\alpha.$$

The detailed structure of each $\mathcal{M}_\alpha$ depends heavily on the *rank* of α , defined as follows:

Definition 4.8. For each $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$, let $K(\alpha)$ be the set of all $k \in H_1(\mathbb{T}^n, \mathbb{Z})$ such that $I(\alpha, k) = 0$, where

$$I : H_{n-1}(\mathbb{T}^n, \mathbb{R}) \times H_1(\mathbb{T}^n, \mathbb{R}) \rightarrow \mathbb{R}$$

is the homology intersection pairing:

$$I(\alpha, k) = \text{PD}(\alpha)(k),$$

with $\text{PD}(\alpha) \in H^1(\mathbb{T}^n, \mathbb{R})$ being the Poincaré dual of α . For the torus, this is just the dot product.

Let $V(\alpha)$ be the smallest subspace of $H_{n-1}(\mathbb{T}^n, \mathbb{R})$ that is spanned by the integer classes and contains α . As in [4], this is equivalent to

$$V(\alpha) = \{\beta \in H_{n-1}(\mathbb{T}^n, \mathbb{R}) : I(\beta, k) = 0 \text{ for all } k \in K(\alpha)\}.$$

We then define $\text{rank}_{\mathbb{Q}}(\alpha) := \dim V(\alpha)$. If $\text{rank}_{\mathbb{Q}}(\alpha) = 1$, we say that α is *rational*. Otherwise, α is called *irrational*. In the extreme case where $\text{rank}_{\mathbb{Q}}(\alpha) = n$, α is said to be *rationally independent* or *totally irrational*.

Note that α is rational if and only if it is a scalar multiple of a class in $H_{n-1}(\mathbb{T}^n, \mathbb{Z})$.

Definition 4.9. A collection of minimizers F with connected supports is called a *lamination* if:

- (1) $\Sigma_1 \in F$ and $\Sigma_2 \in F$ implies $\Sigma_1 \cap \Sigma_2 = \emptyset$ or $\Sigma_1 = \Sigma_2$;
- (2) $\cup_{\Sigma \in F} \Sigma$ is a nonempty, closed subset of $\mathbb{R}^n$.

The lamination F is called a *foliation* if

$$\bigsqcup_{\Sigma \in F} \Sigma = \mathbb{R}^n.$$

The connected components of $\mathbb{R}^n \setminus \cup_{\Sigma \in F} \Sigma$ are called the *gaps* of F . Each gap of F is an open set of $\mathbb{R}^n$, and each lamination contains at most countably many gaps.

For example, for every area-minimizing hypersurface $S \in \mathcal{M}$, the closure of the set $\{\tau_k(S) : k \in \pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n\}$ is a lamination of $\mathbb{R}^n$.

The following structure theorem is due to Bangert in [10] and [7] (for the non-parametric case) and Junginger-Gestrich ([18]) (in the parametric case).

Theorem 4.10. *For every minimizing hypersurface $S \in \mathcal{M}$, there exist a positive integer t and a collection of homology classes $\{\alpha_1, \alpha_2, \dots, \alpha_t\}$ canonically associated to S , defined as follows:*

- (1) *Set $\alpha_1 = \alpha$ and $\Gamma_1 = \mathbb{Z}^n$.*
- (2) *Define inductively the sub-lattices $\Gamma_{s+1} = \alpha_s^\perp \cap \Gamma_s$, provided that this intersection is not empty. If S is not Γ_{s+1} -periodic, then there exists $\alpha_{s+1} \in \text{span}(\Gamma_{s+1})$ such that for all $k \in \Gamma_{s+1}$, properties (4.1) and (4.2) hold. If S is Γ_{s+1} -periodic, this procedure terminates.*

The integer $t = t(S)$ and the set of classes $\alpha_1, \dots, \alpha_t$ (unique up to multiplying by a scalar) are invariants of S , and we write

$$\mathcal{M}(\alpha_1, \dots, \alpha_t) := \{S \in \mathcal{M} : t(S) = t, \alpha_i(S) = \alpha_i, \forall 1 \leq i \leq t\}.$$

Furthermore, for every such $\alpha_1, \alpha_2, \dots, \alpha_t$, the following union

$$\mathcal{M}(\alpha_1) \cup \mathcal{M}(\alpha_1, \alpha_2) \cup \dots \cup \mathcal{M}(\alpha_1, \alpha_2, \dots, \alpha_t) \subset \mathcal{M}_{\alpha_1}$$

is pairwise disjoint and totally ordered (they form a lamination of $\mathbb{R}^n$). In particular, if α is totally irrational, $\mathcal{M}_\alpha$ is totally ordered.

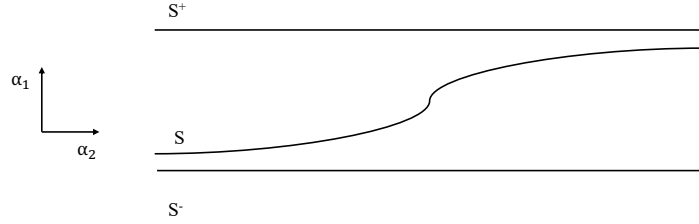


FIGURE 2. When there is a gap bounded by $S^-, S^+ \in \mathcal{M}(\alpha_1)$ that contains no other element in $\mathcal{M}(\alpha_1)$, for any $\alpha_2 \in \alpha_1^\perp \cap \mathbb{Z}^n$, there exists a heteroclinic minimizer $S \in \mathcal{M}(\alpha_1, \alpha_2)$ inside the gap. S is periodic in integral direction orthogonal to both α_1 and α_2 , asymptotic to S^- in the negative α_2 direction and asymptotic to S^+ in the positive α_2 direction.

Remark 4.11. The area-minimizing hypersurfaces in $\mathcal{M}(\alpha_1, \dots, \alpha_t)$ with $t > 1$ are called *heteroclinic minimizers*. They are constructed inside the gaps formed by two consecutive elements in $\mathcal{M}(\alpha_1, \dots, \alpha_{t-1})$ (see Figure 2 above).

It is helpful to keep in mind that the elements in $\mathcal{M}(\alpha_1, \dots, \alpha_{t-1})$ have higher periodicity than the elements in $\mathcal{M}(\alpha_1, \dots, \alpha_t)$, with $\mathcal{M}(\alpha)$ consisting of the most periodic element in $\mathcal{M}_\alpha$. When α is an integer class, $\mathcal{M}(\alpha)$ consists of all the lifts of all closed area-minimizing hypersurfaces in α . If α is totally irrational, we have $\mathcal{M}(\alpha) = \mathcal{M}_\alpha$. In other cases, generically, the inclusion is strict.

5. LAMINATIONS IN AN IRRATIONAL CLASS

In this section, we study area-minimizing hypersurfaces with irrational directions. We begin by establishing the well-order property for $\mathcal{M}_\alpha$, in the case where α is irrational. This result is essentially contained in [6]. The analogue in the non-parametric setting for an elliptic integral was proved earlier by Bangert in [7].

Since the metric g is invariant under integer translations, the actions of elements in $\mathbb{Z}^n$ preserve the area-minimizing property. The fact that the homological vector is preserved is immediate. Therefore, $\mathbb{Z}^n$ acts on each set $\mathcal{M}_\alpha$. When the homology class α is irrational, we denote by $\mathcal{M}_\alpha^{rec}$ the subset of $\mathcal{M}_\alpha$ consisting of minimizing hypersurfaces that can be approximated by their own translates. More precisely, we say that an element $\Sigma \in \mathcal{M}_\alpha$ can be approximated from above if

$$\Sigma = \inf_{k \in \mathbb{Z}^n} \{\tau_k \Sigma : I(k, \alpha) > 0\} \quad (5.1)$$

and from below if

$$\Sigma = \sup_{k \in \mathbb{Z}^n} \{\tau_k \Sigma : I(k, \alpha) < 0\}. \quad (5.2)$$

We now show that the set of recurrent minimizers is well-ordered:

Proposition 5.1. *For irrational α , any two distinct elements in $\mathcal{M}_\alpha^{rec}$ are disjoint.*

We first introduce some notation. For $p \in \mathbb{R}^n$ and $r > 0$, let $Z(p, r)$ be the cylinder with axis $p + \mathbb{R}\alpha$ and radius r . Let $P = \alpha^\perp$ be the hyperplane orthogonal to α . For every ball $B(q, r) \subset \mathbb{R}^n$, let $B_P(q, r) = B(q, r) \cap P$.

Denote by $\mathcal{R}_{n-1}^{loc}(\mathbb{R}^n)$ the space of integral rectifiable currents of locally finite mass on $\mathbb{R}^n$. For $S \in \mathcal{R}_{n-1}^{loc}(\mathbb{R}^n)$ and $W \subset \mathbb{R}^n$ a bounded open set, define the following:

$$\begin{aligned} D(S, W) &:= \sup\{\mathbf{M}(S \llcorner W) - \mathbf{M}(R) : R \in \mathcal{R}_{n-1}^{loc}(\mathbb{R}^n), \partial R = \partial(S \llcorner W)\} \\ &= \mathbf{M}(S \llcorner W) - \inf\{\mathbf{M}(R \llcorner W) : R \in \mathcal{R}_{n-1}^{loc}(\mathbb{R}^n), \partial R = \partial(S \llcorner W)\}, \end{aligned} \quad (5.3)$$

which measures the difference between the mass of $S \llcorner W$ and the mass of the Plateau solution filling $\partial(S \llcorner W)$. It follows from a standard geometric measure theory argument that the function

$$S \rightarrow D(S, W)$$

is mass lower-semicontinuous with respect to the flat topology on $\mathcal{R}_{n-1}^{loc}(\mathbb{R}^n)$.

This quantity is monotonically increasing in the following sense:

Lemma 5.2. *If $W_1 \subseteq W_2$, then*

$$D(S, W_1) \leq D(S, W_2).$$

The proof follows by modifying every competitor of S inside W_1 to get a competitor of S inside W_2 . We omit it here.

Fix $S_1 = \partial E_1$ and $S_2 = \partial E_2$ in $\mathcal{M}_\alpha^{rec}$. Assume, for contradiction, that $S_1 \neq S_2$ and they intersect. We can assume further that both of them can be approximated from below as follows. Since they are recurrent, there exist two sequences of recurrent minimizers $S_1^i \rightarrow S_1$ and $S_2^i \rightarrow S_2$ such that every S_1^i and S_2^i can be approximated from below. Since $S_1 \neq S_2$ and they intersect, it follows that for sufficiently large i , we have $S_1^i \neq S_2^i$ and $S_1^i \cap S_2^i \neq \emptyset$. We can thus work with S_1^i and S_2^i instead of S_1 and S_2 .

Define the following:

$$\begin{aligned} S^+ &= \max(S_1, S_2) = \partial(E_1 \cup E_2), \\ S^- &= \min(S_1, S_2) = \partial(E_1 \cap E_2). \end{aligned}$$

Lemma 5.3. *There exist $\epsilon > 0$ and $\rho > 0$ such that for all $p \in P$ and $r \geq \rho$, we have*

$$D(S^+, B(p, r)) \geq \epsilon \text{ and } D(S^-, B(p, r)) \geq \epsilon.$$

Proof. We prove the statement for S^+ by contradiction; the one for S^- is completely analogous. Assume the contrary. Then we can find a sequence of points $p_i \in P$ and radii $r_i \rightarrow \infty$ such that

$$D(S^+, B(p_i, r_i)) \rightarrow 0.$$

There exists a corresponding sequence of integer translations $\tau_{k_i} \in \pi_1(\mathbb{T}^n)$ such that

$$q_i = \tau_{k_i}(p_i) = p_i + k_i \in [0, 1]^n.$$

This choice of translations ensures that the distance from q_i to the plane P is uniformly bounded. As a result, for $j = 1, 2$, elements of the sequence $\{\tau_{k_i} S_j\}_i$ always lie within a bounded distance from P . By the compactness property of the minimizers (Theorem 4.7), we have $\tau_{k_i} S_j \rightarrow \tilde{S}_j = \partial \tilde{E}_j$ for $j = 1, 2$. This means that

$$\tau_{k_i} S^+ \rightarrow \tilde{S}^+ = \partial(\tilde{E}_1 \cup \tilde{E}_2).$$

Therefore

$$\lim_{i \rightarrow \infty} D(\tau_{k_i} S^+, B(q_i, r_i)) = \lim_{i \rightarrow \infty} D(S^+, B(p_i, r_i)) = 0.$$

For every $r > 0$, we can choose i sufficiently large so that $B(0, r) \subset B(q_i, r_i)$. It follows from the monotonicity lemma 5.2 that

$$D(\tau_{k_i} S^+, B(0, r)) \rightarrow 0.$$

From the lower-semicontinuity of the mass, we get $D(\tilde{S}^+, B(0, r)) = 0$ for all r . Therefore $\tilde{S}^+$ is area-minimizing.

Note that $\tilde{E}_1 \subset \tilde{E}_1 \cup \tilde{E}_2$. Since both $\tilde{S}_1$ and $\tilde{S}^+$ are area-minimizing, it follows from the maximum principle (cf. [43]) that $\tilde{S}_1 = \tilde{S}^+$ or $\tilde{S}_1 \cap \tilde{S}^+ = \emptyset$. Therefore, $\tilde{S}_1 = \tilde{S}_2$ or $\tilde{S}_1 \cap \tilde{S}_2 = \emptyset$. In both cases we have $\tilde{S}_2 \leq \tilde{S}_1$. Since

$S_1 \in \mathcal{M}_\alpha^{rec}$ can be approximated from below, we can find a sequence of integer translations $\{\tau_{h_i}\}_i$ such that

$$\tau_{h_i}\tilde{S}_1 \leq S_1, \quad \forall i \in \mathbb{N} \quad \text{and} \quad \tau_{h_i}\tilde{S}_1 \rightarrow S_1.$$

By Lemma 4.7 in [7], we have that $\tau_{h_i}\tilde{S}_2 \rightarrow S_2$. Since $\tilde{S}_2 \leq \tilde{S}_1$, it follows that

$$\tau_{h_i}\tilde{S}_2 \leq \tau_{h_i}\tilde{S}_1, \quad \forall i \in \mathbb{N}.$$

Consequently $S_2 \leq S_1$. We get a contradiction. $\square$

Lemma 5.4. *There exist $\delta > 0$ and $r_0 > 0$ such that*

$$D(S^+, Z(0, r)) > \delta r^{n-1} \quad \text{and} \quad D(S^-, Z(0, r)) > \delta r^{n-1}$$

for all $r > r_0$.

Proof. Fix the constant ρ as in Lemma 5.3. There exists a positive constant $c > 0$ such that, for all sufficiently large r , the ball $B_P(0, r)$ contains at least $N(r) := cr^{n-1}$ pairwise disjoint open balls of radius ρ , which we shall denote by $B_P(p_i, \rho)$ for $i = 1, 2, \dots, N(r)$. It follows that the Euclidean balls $B(p_i, \rho)$ are also disjoint and contained in the cylinder $Z(0, r)$ for all i . Therefore

$$D(S^+, Z(0, r)) \geq \sum_{i=1}^{N(r)} D(S^+, Z(p_i, \rho)) \geq \epsilon cr^{n-1}.$$

The second inequality follows from Lemma 5.3 and the fact that the balls are contained inside the cylinders. The desired inequality follows by choosing $\delta = \epsilon c$. $\square$

We can now complete the proof of Proposition 5.1. By Theorem 4.1, we can choose a slab of bounded width B that contains all points of bounded distance C to P . In particular, B contains both S_1 and S_2 .

Choose r sufficiently large. By Lemma 5.4, we can find competitors T_r^+, T_r^- with $\partial T_r^\pm = \partial(S^\pm \llcorner Z(0, r))$ such that

$$\mathbf{M}(T_r^+) \leq \mathbf{M}(S^+ \llcorner Z(0, r)) - \delta r^{n-1} \quad \text{and}$$

$$\mathbf{M}(T_r^-) \leq \mathbf{M}(S^- \llcorner Z(0, r)) - \delta r^{n-1}.$$

Adding these two inequalities, we get

$$\begin{aligned} \mathbf{M}(T_r^+) + \mathbf{M}(T_r^-) &\leq \mathbf{M}(S^+ \llcorner Z(0, r)) + \mathbf{M}(S^- \llcorner Z(0, r)) - 2\delta r^{n-1} \\ &= \mathbf{M}(S_1 \llcorner Z(0, r)) + \mathbf{M}(S_2 \llcorner Z(0, r)) - 2\delta r^{n-1}. \end{aligned} \tag{5.4}$$

Since S_1 and S_2 are area-minimizing and belong to the bounded slab B , their masses satisfy

$$\mathbf{M}(S_1 \llcorner Z(0, r)) \leq \mathbf{M}(T_r^+) + \mathbf{M}(\partial(Z(0, r) \llcorner B)) \leq \mathbf{M}(T_r^+) + Cr^{n-2}$$

for some constant $C > 0$. Similarly,

$$\mathbf{M}(S_2 \llcorner Z(0, r)) \leq \mathbf{M}(T_r^-) + Cr^{n-2}.$$

Combining these inequalities with (5.4), we obtain

$$\mathbf{M}(T_r^+) + \mathbf{M}(T_r^-) \leq \mathbf{M}(T_r^+) + \mathbf{M}(T_r^-) + 2Cr^{n-2} - 2\delta r^{n-1}.$$

Rearranging gives

$$\delta r^{n-1} \leq Cr^{n-2},$$

which is impossible when r is sufficiently large. We reach a contradiction, and this completes the proof of Proposition 5.1. $\square$

Remark 5.5. A variant of the preceding argument yields an even stronger conclusion in the case where α is totally irrational: any two distinct elements in $\mathcal{M}_\alpha$ are disjoint. In other words, $\mathcal{M}_\alpha$ is totally ordered. In addition to that, Bangert (cf. [7]) proved the following stronger uniqueness result: the set $\mathcal{M}_\alpha^{rec}$ is the unique minimal set of $\mathcal{M}_\alpha$, under the action of the group of integer translations. This gives a dynamical characterization of the subset consisting of recurrent minimizers.

It is well-known that when α is irrational, the set of all recurrent minimizers $\mathcal{M}_\alpha^{rec}$ either forms a foliation of $\mathbb{R}^n$ or is homeomorphic to a Cantor set in $\mathbb{R}$. In the latter case, the open intervals belonging to the complement of the Cantor set are called the gaps of the lamination, with its endpoints corresponding to elements in $\mathcal{M}_\alpha^{rec}$ that can only be approximated from above or below, i.e., only one of (5.1) or (5.2) holds. The remaining uncountably many other minimizers are those that can be approximated both from above and from below.

The following lemma provides a universal finite upper bound for the volumes of all such gaps in the lamination.

Lemma 5.6. *Suppose that α is totally irrational and G is a gap in the lamination generated by recurrent minimizers in $\mathcal{M}_\alpha$. Then the projection $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$ is injective when restricted to G . Consequently,*

$$\text{Vol}_g(G) \leq \text{Vol}_g(\mathbb{T}^n) < \infty.$$

Proof. Let $\Sigma^- < \Sigma^+$ be elements in $\mathcal{M}_\alpha^{rec}$ that form the boundary of G . Since G is a gap of the lamination, $\Sigma^\pm$ are consecutive, meaning that no other recurrent minimizers can be found between them.

Recall that the set $\mathcal{M}_\alpha^{rec}$ is totally ordered and invariant under integer translations. As a result, for every translation $\tau_k \in \mathbb{Z}^n$, $\tau_k(\Sigma^\pm)$ are again recurrent minimizers. Furthermore, since α is totally irrational, for any nonzero $k \in \mathbb{Z}^n$ we have $I(\alpha, k) \neq 0$. It follows from the Birkhoff property that $\Sigma^\pm$ and $\tau_k(\Sigma^\pm)$ are disjoint.

Therefore, $\tau_k(\Sigma^\pm)$ bound the translated gap $\tau_k(G)$, which implies that G and $\tau_k(G)$ are disjoint. Since this is true for every integer translation, it follows that the projection $\pi|_G$ is injective. This completes the proof. $\square$

Remark 5.7. In general, for a prescribed irrational class α , there exist metrics on $\mathbb{T}^n$ for which non-recurrent minimizers do appear. The following construction of such metrics is due to Bangert in [8].

Fix a metric g_0 on $\mathbb{T}^n$ and assume that every minimizer in $\mathcal{M}_\alpha(g_0)$ is recurrent. We may further assume that $\mathcal{M}_\alpha(g_0) = \mathcal{M}_\alpha^{rec}(g_0)$ does not form a foliation, otherwise we are done. Let G be one of the gaps, and choose a compact set K contained inside G . We then decrease the metric g_0 inside K to obtain a new metric g_1 , which agrees with g_0 outside K , and for which the original minimizers with respect to g_0 are no longer minimizers with respect to g_1 .

Consider the following family of interpolated metrics:

$$g_t = (1 - t)g_0 + tg_1, \quad t \in [0, 1].$$

These are the affine combinations of g_0 and g_1 . By the compactness property of minimizers, there exists a maximal time $0 < T < 1$ such that the original minimizers in $\mathcal{M}_\alpha(g_0)$ remain minimizers with respect to g_T . By the definition of recurrent minimizers, it follows that

$$\mathcal{M}_\alpha^{rec}(g_T) = \mathcal{M}_\alpha^{rec}(g_0).$$

For every $t > T$, there must exist a minimizer $\Sigma_t \in \mathcal{M}_\alpha(g_t)$ that intersects K . Taking the limit as $t \rightarrow T^+$, by the compactness property of area-minimizing hypersurfaces, the sequence $\{\Sigma_t\}$ converges to a minimizer $\Sigma_T \in \mathcal{M}_\alpha(g_T)$ that intersects K . This minimizer is non-recurrent.

This phenomenon, however, is not generic. In fact, an arbitrarily small perturbation of the metric suffices to eliminate all non-recurrent minimizers in $\mathcal{M}_\alpha$. Equivalently, $\mathcal{M}_\alpha = \mathcal{M}_\alpha^{rec}$ generically. Nonetheless, recurrent minimizing hypersurfaces are limits of periodic minimizers.

Proposition 5.8 (Bangert, [7]). *For any recurrent minimizer $\Sigma \in \mathcal{M}_\alpha^{rec}$ and any sequence of integer classes r_k converging to α , there exists a subsequence (still denoted by r_k) and $\Sigma_k \in \mathcal{M}_{r_k}^{per}$ such that $\Sigma_k \rightarrow \Sigma$.*

Proof. For each k , fix a periodic minimizer $R_k \in \mathcal{M}_{r_k}^{per}$, then consider the discrete set

$$\mathcal{O}_k = \{\tau_s R_k : s \in \pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n\}.$$

Let $\mathcal{O}$ be the closure of the union of all $\mathcal{O}_k$. Choose a sequence of minimizers $S_k \in \mathcal{O}_k$ intersecting a fixed compact set. By Theorem 4.7, we have

$$S_k \rightarrow S \in \mathcal{M}_\alpha \cap \mathcal{O}.$$

Since the metric is periodic, $\mathcal{O}$ is closed and invariant under integer translations. Therefore $\overline{\{\tau_s S : s \in \mathbb{Z}^n\}} \subseteq \mathcal{O}$. Since $\mathcal{M}_\alpha^{rec}$ is the unique minimal set of $\mathcal{M}_\alpha$ under the action of integer translations, we conclude that

$$\mathcal{M}_\alpha^{rec} \subseteq \overline{\{\tau_s S : s \in \mathbb{Z}^n\}} \subseteq \mathcal{O}.$$

This completes the proof of the proposition. $\square$

It was conjectured by Moser [35] that the above inclusions should be equalities, meaning all limits of sequences of periodic minimizers are recurrent. This is not known to be true yet. The following proposition gives an obstruction for the alternative in terms of the number of closed stable hypersurfaces in a sequence of integer classes approaching α .

Proposition 5.9. *Let $r_k \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ be a sequence of primitive integer classes converging to a totally irrational class α . Suppose that there exists periodic minimizers $\Sigma_k \in \mathcal{M}_{r_k}^{per}$ such that*

$$\Sigma_k \rightarrow \Sigma \in \mathcal{M}_\alpha \setminus \mathcal{M}_\alpha^{rec}$$

locally smoothly in $\mathbb{R}^n$. Assume further that Σ is isolated in $\mathcal{M}_\alpha$. Then there exists $K \in \mathbb{N}$ such that for all $k \geq K$, there exists another connected stable minimal hypersurface $\tilde{\Sigma}_k$ different from $\pi(\Sigma_k)$, representing the class r_k . In particular, if for every positive integer k , $\pi(\Sigma_k)$ is the unique connected stable minimal hypersurface representing the class r_k , then Σ is recurrent or a non-isolated non-recurrent minimizer in $\mathcal{M}_\alpha$.

Proof. Since Σ is not recurrent, Bangert [7] showed that Σ lies in a gap G bounded by two recurrent minimizers $\Sigma^- < \Sigma^+$ in $\mathcal{M}_\alpha^{rec}(g)$, where

$$\Sigma^+ = \inf\{\tau_s \Sigma : s \in \pi_1(\mathbb{T}^n), s \cdot \alpha > 0\},$$

and

$$\Sigma^- = \sup\{\tau_s \Sigma : s \in \pi_1(\mathbb{T}^n), s \cdot \alpha < 0\}.$$

By Lemma 5.6, the projection $\pi : G \rightarrow \mathbb{T}^n$ is injective, and G has finite g -volume. Choose a point $p \in \Sigma$. Since Σ is isolated in $\mathcal{M}_\alpha(g)$, we may choose a relatively compact open set $\Omega \subset G$ with smooth boundary such that $p \in \Omega$, $\Omega \cap \Sigma \neq \emptyset$, and no element of $\mathcal{M}_\alpha(g)$ other than Σ intersects $\overline{\Omega}$.

Let $\varphi \in C^\infty(\mathbb{T}^n)$ be nonnegative, supported in $\pi(\Omega)$, and positive on $\pi(\Omega)$. This lifts to a $\mathbb{Z}^n$ -periodic function on $\mathbb{R}^n$. Define a new metric

$$g_\varphi := e^{2\varphi} g \quad \text{on } \mathbb{T}^n.$$

The lifted metric on $\mathbb{R}^n$ will still be denoted by g_φ . By construction, $g_\varphi \geq g$, and $g_\varphi = g$ outside $\pi(\Omega)$. Since $\Sigma^\pm$ are disjoint from the support of the perturbation, they remain area-minimizing with respect to g_φ . Indeed, for every compactly supported competitor T :

$$\mathbf{M}_{g_\varphi}(T \llcorner W) \geq \mathbf{M}_g(T \llcorner W) \geq \mathbf{M}_g(\Sigma^\pm \llcorner W) = \mathbf{M}_{g_\varphi}(\Sigma^\pm \llcorner W).$$

We claim that no element of $\mathcal{M}_\alpha(g_\varphi)$ intersects the region $\overline{\Omega}$. Suppose, to the contrary, that $\Gamma \in \mathcal{M}_\alpha(g_\varphi)$ intersects this region. If Γ does not meet Ω , then it must intersect $\partial\Omega$ and belong to $\mathcal{M}_\alpha(g)$, contradicting the assumption that no other minimizer in $\mathcal{M}_\alpha(g)$ intersects $\overline{\Omega}$. Hence we can assume $\Gamma \cap \Omega \neq \emptyset$.

Since $\Sigma^\pm$ are g_φ -minimizing and have the same homological direction as Γ , the Birkhoff property implies that Γ lies between Σ^- and Σ^+ , hence inside the same gap G . Let $B_R = B_R(p)$, where the radius is taken with respect to the auxiliary proper metric defined in Section 6.

Since G has finite volume, Lemma 6.3 gives

$$\text{Vol}_{n-1}^g(\partial B_R \cap G) \rightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Consequently, the restrictions of Γ and $\Sigma^\pm$ to B_R can be compared after adding a filling current supported in $\partial B_R \cap G$, whose mass tends to zero as

$R \rightarrow \infty$. In particular, since $\Sigma^\pm$ are g -area-minimizing, there exists an error term $\eta_R \rightarrow 0$ such that

$$\mathbf{M}_g(\Gamma \llcorner B_R) \geq \mathbf{M}_g(\Sigma^\pm \llcorner B_R) - \eta_R.$$

On the other hand, because Γ meets the region where $\varphi > 0$, there exists $\lambda > 0$ such that, for all sufficiently large R , we have

$$\mathbf{M}_{g_\varphi}(\Gamma \llcorner B_R) \geq \mathbf{M}_g(\Gamma \llcorner B_R) + 2\lambda.$$

Combining the preceding two estimates and using $g_\varphi = g$ outside $\pi(\Omega)$, we obtain

$$\mathbf{M}_{g_\varphi}(\Gamma \llcorner B_R) \geq \mathbf{M}_{g_\varphi}(\Sigma^\pm \llcorner B_R) + 2\lambda - \eta_R.$$

For R sufficiently large, this is strictly larger than $\mathbf{M}_{g_\varphi}(\Sigma^\pm \llcorner B_R) + \lambda$, since $\eta_R \rightarrow 0$. However, the same small filling current on $\partial B_R \cap G$ gives an admissible competitor for $\Gamma \llcorner B_R$ whose g_φ -mass is at most $\mathbf{M}_{g_\varphi}(\Sigma^\pm \llcorner B_R) + \eta_R$. This contradicts the fact that Γ is area-minimizing with respect to g_φ and proves the claim.

Now, for each k , let $\tilde{\Sigma}_k$ be a closed area-minimizing hypersurface in the class r_k with respect to g_φ , and choose a lift

$$\Sigma'_k \in \mathcal{M}_{r_k}^{per}(g_\varphi).$$

We argue that, for all sufficiently large k , no lift of $\tilde{\Sigma}_k$ intersects Ω . Otherwise, after passing to a subsequence and choosing appropriate integer translates, we would obtain a sequence Σ'_k intersecting Ω . By the compactness theorem 4.7, a subsequence converges locally smoothly to $\Gamma \in \mathcal{M}_\alpha(g_\varphi)$ which intersects $\bar{\Omega}$, contradicting the claim. Therefore, for all sufficiently large k , the hypersurface $\tilde{\Sigma}_k$ is disjoint from Ω . Hence $\tilde{\Sigma}_k$ is not only minimal and stable with respect to g_φ , but also with respect to the original metric g . Since r_k is primitive, the minimizing representative $\tilde{\Sigma}_k$ is connected.

Finally, because $\Sigma_k \rightarrow \Sigma$ and Σ intersects the region where $\varphi > 0$, the projected hypersurface $\pi(\Sigma_k)$ also intersects this region for all sufficiently large k . By contrast, $\tilde{\Sigma}_k$ is disjoint from it. Hence $\tilde{\Sigma}_k \neq \pi(\Sigma_k)$. This completes the proof. $\square$

6. NECESSARY CONDITION FOR MINIMAL FOLIATION

In this section, we will prove the implication (3) $\rightarrow$ (1) in the statement of Theorem 1.3, together with Theorem 1.6. We restate them here for convenience.

Theorem 6.1. *Suppose that $\mathcal{M}_\alpha$ foliates $\mathbb{R}^n$ for some totally irrational $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$. Then*

$$\Delta W_\alpha = \lim_{k \rightarrow \infty} (\omega(r_k) - S(r_k)) = 0$$

for any sequence of integer classes $\{r_k\}_k$ converging to α . If α is rationally dependent, the same conclusion holds whenever $\mathcal{M}(\alpha)$ foliates $\mathbb{R}^n$.

Proof. The first equality automatically holds from the definition of the Mather energy barrier once we establish the second equality for every sequence of homology classes $\{r_k\}_k \rightarrow \alpha$.

We begin with the case in which α is totally irrational. Without loss of generality, assume that there is no foliation of $\mathbb{R}^n$ by closed, minimal hypersurfaces in the homology class r_k for all $k \in \mathbb{N}$ (if such a foliation exists, then $\omega(r_k) - S(r_k) = 0$, so we could simply discard those r_k from the original sequence without affecting the zero limit). For any k , let $M_k < N_k$ be two consecutive elements in $\mathcal{M}(r_k)$ that are lifts of the same homologically area-minimizing element in the class r_k . Then $N_k = \tau_{t_k} M_k$ for some integer translation τ_{t_k} . We will work on their restrictions to a fundamental domain in $\mathbb{R}^n$.

For a fixed $\epsilon > 0$, we want to construct a one-parameter family Φ_t interpolating from M_k to N_k such that

$$\mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t) \leq \mathbf{M}(M_k) + \epsilon$$

for all t , when $k > K$ is sufficiently large. Here, the masses are of the compact projections of M_k and N_k to $\mathbb{T}^n$ or of their restrictions to a fundamental domain on $\mathbb{R}^n$. We shall not distinguish between these two when no confusion can arise.

We proceed as follows. By applying integer translations, we can assume that M_k and N_k always intersect the unit cube for all k . It follows from the compactness property of the set of minimizers $\mathcal{M}$ that

$$M_k, N_k \longrightarrow M, N \in \mathcal{M}_\alpha.$$

We first treat the case where the limits M and N are distinct. Since $\mathcal{M}_\alpha$ is a foliation, the region between M and N is foliated by area-minimizing hypersurfaces of direction α . Let H_k be the fundamental domain bounded by M_k and N_k . The idea is to partition H_k into two collections of open balls.

- (1) **Type A:** balls on which M_k and N_k are C^1 -close.
- (2) **Type B:** balls on which M_k and N_k are not sufficiently close.

On an open ball U of type A, we can use the interpolation lemma to connect $M_k \llcorner U$ to $N_k \llcorner U$, since they are area-minimizing and close in the flat norm. On an open ball W of type B, we connect $M_k \llcorner W$ to $N_k \llcorner W$ by modifying the foliation in $\mathcal{M}_\alpha$.

Assume that G is the open set bounded by M and N .

Claim 6.2. *The restriction of the projection $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$ to the gap G is injective.*

Proof of claim. We prove this claim by contradiction. Suppose that $\pi(x) = \pi(y)$ for some distinct points $x, y \in G$. Choose a large compact set K such that $x, y \in K \cap G$. Since M_k and N_k converge to M and N on compact sets, we have

$$M_k \llcorner K \rightarrow M \llcorner K \quad \text{and} \quad N_k \llcorner K \rightarrow N \llcorner K.$$

Furthermore, since α is totally irrational and $r_k \rightarrow \alpha$, the fundamental domains H_k expand along every direction. Therefore, for a sufficiently large k , we have

$x, y \in H_k$. This contradicts the bijectivity of π on H_k and therefore proves the claim. $\square$

Let $d_g(x) = \text{dist}_g(x, 0)$ be the distance function with respect to the periodic metric g . We can choose a smooth approximation $\tilde{d}$ of d such that

$$\sup_{x \in \mathbb{R}^n} |\tilde{d}(x) - d(x)| \leq 1$$

and

$$\text{Lip}(\tilde{d}) \leq 2.$$

Let $B_r = \{x \in \mathbb{R}^n | \tilde{d}(x) \leq r\}$. By the co-area formula, we have

$$\int_0^\infty \text{Vol}_{n-1}(G \cap \partial B_r) dr \leq \text{Lip}(\tilde{d}) \text{Vol}_n(G) < \infty.$$

Since M and N are area-minimizing, it follows from the standard curvature estimates for stable minimal hypersurfaces that outside of a sufficiently large compact set, one can write M locally as a normal graph over N :

$$M = \{\exp_x(u(x)\nu(x)) : x \in N\}$$

where $u > 0$ satisfies a uniformly elliptic equation. A standard argument using Harnack's inequality yields the following decay lemma, which is useful in applying an area comparison argument later. We postpone the proof of it to Appendix D.

Lemma 6.3. *The following holds:*

$$\lim_{R \rightarrow \infty} \text{Vol}_{n-1}(\partial B_R(0) \cap G) = 0.$$

Fix a small constant $\eta > 0$. Since M_k and N_k are stable, for any $x \in M_k$ such that $\text{dist}(x, N_k) > 2\eta$, a standard argument using curvature estimates for stable minimal hypersurfaces shows that there exists δ_η such that for all

$$y \in B(x, \delta_\eta) \cap M_k,$$

we have $\text{dist}(y, N_k) > \eta$. Thus we can choose a collection Γ_k of at most N_η balls $B \subset H_k$ such that the following properties hold:

- (1) The diameters of all $B \in \Gamma_k$ are uniformly bounded independent of k .
- (2) $\mathbf{M}(M_k \llcorner B) = \mathbf{M}(N_k \llcorner B)$.
- (3) For any $y \in M_k \setminus \cup_{B \in \Gamma_k} B$, the distance from y to N_k is at most η .

Step 1: Interpolating from M_k to N_k inside each ball $B \in \Gamma_k$.

We now work on the restriction $M_k \llcorner B, N_k \llcorner B$ for each $B \in \Gamma_k$. Since B is uniformly bounded, by translation, we can assume that

$$M_k \llcorner B, N_k \llcorner B \subset \Omega$$

for some fixed bounded compact set $\Omega \in \mathbb{R}^n$. Since M_k and N_k converge to M and N on compact sets, we have

$$M_k \llcorner B \rightarrow M \llcorner B \text{ and } N_k \llcorner B \rightarrow N \llcorner B.$$

Since $\mathcal{M}_\alpha$ foliates $\mathbb{R}^n$, we can find a map

$$\varphi_t : [0, 1] \rightarrow \mathcal{M}_\alpha, \quad \varphi_0 = M, \quad \varphi_1 = N.$$

Let $\varphi_t = \partial E_t$ and $M_k = \partial E$. Consider the current:

$$\Phi_t = M_k + \partial((E_t \setminus E) \llcorner B).$$

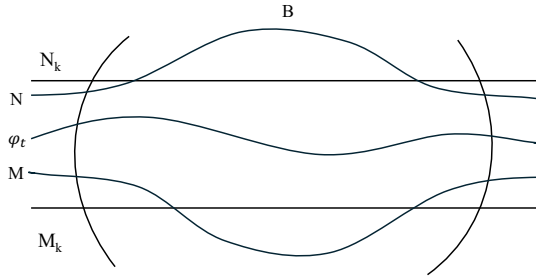


FIGURE 3. The local interpolation from M_k to N_k inside B is constructed from the foliation in $\mathcal{M}_\alpha$.

We will interpolate from M_k to $M_k + \partial(B \cap H_k)$ in the following order:

$$M_k \rightarrow M_k + \partial((E_0 \setminus E) \cap B) \rightarrow M_k + \partial((E_1 \setminus E) \cap B) \rightarrow M_k + \partial(B \cap H_k).$$

Let $\epsilon' > 0$ be chosen later. Since $M_k, N_k \rightarrow M, N$ on Ω , for sufficiently large k , we can interpolate from M_k to $M_k + \partial((E_0 \setminus E) \cap B)$ and from $M_k + \partial((E_1 \setminus E) \cap B)$ to $M_k + \partial(B \cap H_k)$ with mass changes of at most ϵ' .

Note that both M_k and the hypersurfaces φ_t are area-minimizing for all t , and for small η , $\partial(B \cap H_k)$ has small area. It follows from the area comparison argument that:

$$|\mathbf{M}(M_k + \partial((E_t \setminus E) \cap B)) - \mathbf{M}(M_k)| < \epsilon'.$$

Therefore, we can construct a family Φ_t interpolating from M_k to $M_k + \partial(B \cap H_k)$ such that:

$$\mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t) \leq \mathbf{M}(M_k) + 3\epsilon'.$$

We may repeat this construction for every ball $B \in \Gamma_k$.

Since the number of such balls is at most N_η , we get a $\mathcal{F}$ -continuous family $\{\Phi_t^1\}$ joining M_k to $M_k + \partial(\bigcup_{B \in \Gamma_k} B \cap H_k)$ and satisfying the mass estimate

$$\mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t^1) \leq \mathbf{M}(M_k) + 3N_\eta \epsilon'.$$

Choosing ϵ' such that $3N_\eta \epsilon' = \epsilon$, we conclude that every slice Φ_t^1 in the family has mass bounded by $\mathbf{M}(M_k)$ and $\mathbf{M}(M_k) + \epsilon$, for all $k > K_1$. Moreover,

the family exhibits no concentration of mass. This follows from the fact that each slice is obtained by modifying smooth area-minimizing hypersurfaces in $\mathcal{M}_\alpha$ only in a controlled way.

Step 2: Interpolating outside $\bigcup_{B \in \Gamma_k} B$.

We now turn to the problem of interpolating from the current

$$M_k + \partial\left(\bigcup_{B \in \Gamma_k} B \cap H_k\right)$$

to N_k . This current is naturally decomposed into three types of pieces:

- (1): Portions of M_k lying outside of the balls in Γ_k , all of which are η -close (in flat norm and C^1) to the corresponding portions of N_k ;
- (2): Boundary pieces $\partial(B \cap H_k)$ for all $B \in \Gamma_k$, each of which has small area because $B \in \Gamma_k$;
- (3): Pieces of N_k lying inside the union $\bigcup_{B \in \Gamma_k} B$.

Rather than carrying along this decomposition explicitly, we may simplify the discussion by assuming that M_k and N_k are uniformly η -close in C^1 . Under this reduction, the interpolation becomes a local problem on a finite covering of the fundamental domain.

More precisely, we partition the fundamental domain into finitely many open sets:

$$H_k = \sum_{i=1}^{J_k} Q_{k,i},$$

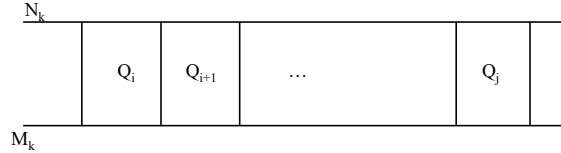


FIGURE 4. A partition of the fundamental domain into sets of small boundaries.

where $\mathbf{M}(\partial Q_{k,i}) < \eta^{n-1}$ is small enough to allow a controlled interpolation between the portions of M_k and N_k lying inside $Q_{k,i}$.

The boundary of each cube $Q_{k,i}$ consists of faces intersecting M_k , N_k , and the sides that belong to the gap. Let $S(Q_{k,i})$ be the sides that belong to the gap.

Since α is totally irrational, we can approximate a fundamental domain of M_k by a rectangular cube whose sides $x_1^{(k)}, x_2^{(k)}, \dots, x_{n-1}^{(k)}$ go to infinity. Since the volume of the fundamental domain H_k is 1, we have $\eta x_1^{(k)} x_2^{(k)} \cdots x_{n-1}^{(k)} \approx 1$

and hence the surface area of H_k excluding the two faces belonging to M_k, N_k is approximately

$$2\eta(x_1^{(k)} + \cdots + x_{n-1}^{(k)}) \approx \frac{2}{x_2^{(k)} \cdots x_{n-1}^{(k)}} + \cdots + \frac{2}{x_1^{(k)} \cdots x_{n-2}^{(k)}} \rightarrow 0.$$

Therefore, we can arrange $Q_{k,i}$ in such a way that:

$$\mathbf{M}(S(\sum_{i=1}^j Q_{k,i})) \leq \eta^{n-1}$$

for all $1 \leq j \leq J_k$. Let $P_{k,j} = \sum_{i=1}^j Q_{k,i}$. Then $P_{k,0} = 0$ and $P_{k,J_k} = H_k$. Since N_k minimizes the area, we have

$$\mathbf{M}(N_k \sqcup P_{k,j}) \leq \mathbf{M}(M_k \sqcup P_{k,j}) + \mathbf{M}(S(P_{k,j})). \quad (6.1)$$

Furthermore

$$M_k + \partial P_{k,j} = (M_k - P_{k,j}) + N_k \sqcup P_{k,j} + S(P_{k,j}),$$

it follows that:

$$\mathbf{M}(M_k + \partial P_{k,j}) \leq \mathbf{M}(M_k - P_{k,j}) + \mathbf{M}(N_k \sqcup P_{k,j}) + \mathbf{M}(S(P_{k,j})). \quad (6.2)$$

Combining (6.1) and (6.2), we have the following.

$$\begin{aligned} \mathbf{M}(M_k + \partial P_{k,j}) &\leq \mathbf{M}(M_k - P_{k,j}) + \mathbf{M}(M_k \sqcup P_{k,j}) + 2\mathbf{M}(S(P_{k,j})) \\ &\leq \mathbf{M}(M_k) + 2\eta^{n-1}. \end{aligned} \quad (6.3)$$

We want to apply Corollary 3.2 to $T_0 = M_k + \partial P_{k,j-1}$ and $T_1 = M_k + \partial P_{k,j}$. Note that

$$\mathbf{M}(T_1 - T_0) = \mathbf{M}((M_k + \partial P_{k,j}) - (M_k + \partial P_{k,j-1})) = \mathbf{M}(\partial Q_{k,j}) < \eta^{n-1}.$$

We now choose η so that

$$\eta^{n-1} < \min \left\{ \delta_0, \frac{\epsilon}{2 + C_0} \right\},$$

with δ_0 and C_0 as chosen in the interpolation lemma, Corollary 3.2. It follows that we can interpolate from $M_k + \partial P_{k,j-1}$ to $M_k + \partial P_{k,j}$ such that the mass of each element is at most

$$\mathbf{M}(M_k + \partial P_{k,j}) + C_0 \eta^{n-1} \leq \mathbf{M}(M_k) + 2\eta^{n-1} + C_0 \eta^{n-1} \leq \mathbf{M}(M_k) + \eta^{n-1}(2 + C_0),$$

with the first inequality following from (6.3). On the other hand, since M_k is homologically area-minimizing in r_k , we see that the mass of each slice is at least the mass of M_k . By the choice of η , we have

$$\eta^{n-1}(2 + C_0) < \epsilon.$$

Therefore, by letting j range from 1 to J_k , we get a mass-continuous family Φ_t^2 of integer currents in $\mathcal{Z}_{n-1}(\mathbb{T}^n, r_k)$ from M_k to $M_k + \partial P_{k,J_k}$ that satisfies

$$\mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t^2) \leq \mathbf{M}(M_k) + \epsilon$$

for all $k > K_2$. Finally, since $M_k + \partial P_{k,J_k} = N_k + S(P_{k,J_k})$ and $M(S(P_{k,J_k}))$ is small, we can interpolate from $M_k + \partial P_{k,J_k}$ to N_k by a mass-continuous family Φ_t^3 such that

$$\mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t^3) \leq \mathbf{M}(M_k) + \epsilon, \quad \forall k > K_3.$$

According to Lemma 2.7 in [25], Φ_t^2 and Φ_t^3 are continuous in the flat topology and have no mass concentration.

As a result, we can combine Φ^1 , Φ^2 , and Φ^3 into a single sweep-out Φ and deduce that

$$S(r_k) = \mathbf{M}(M_k) \leq \mathbf{M}(\Phi_t) \leq \mathbf{M}(M_k) + \epsilon = S(r_k) + \epsilon$$

for all $k > \max\{K_1, K_2, K_3\}$. It follows that $\omega(r_k) \leq S(r_k) + \epsilon$ for all k sufficiently large.

It remains to analyze the case in which $M = N$ in $\mathcal{M}_\alpha$, namely when the sequences $\{M_k\}_k$ and $\{N_k\}_k$ converge to the same limit. We make the following claim.

Claim 6.4. *We can assume the following: for any $\epsilon > 0$, there exists K such that for all $k > K$, we can write N_k locally as a graph of a function f_k over M_k , such that $|f_k|_{C^2} < \epsilon$.*

Proof of claim. We argue by contradiction. Suppose that the claim were false. Then there would exist some $\epsilon > 0$ and a sequence of indices k for which N_k cannot be written, in a neighborhood of certain points, as a graph over M_k with C^2 -norm smaller than ϵ .

Since both M_k and N_k are stable, it follows that we can find a sequence of points $x_k \in M_k$ such that for some open ball B_k containing x_k of fixed radius δ , the volume of the gap between M_k and N_k lying inside B_k is uniformly bounded from below. We can use appropriate integer translations τ_{t_k} that move B_k to a fixed cube $B \in \mathbb{R}^n$ and then consider the new sequences $\widetilde{M}_k = \tau_{t_k} M_k$, $\widetilde{N}_k = \tau_{t_k} N_k$.

Because the volume of the region they bound inside a fixed cube B is uniformly bounded from below away from zero, $\{\widetilde{M}_k\}_k$ and $\{\widetilde{N}_k\}_k$ cannot converge to the same limit. This means that we are in the situation of the case we considered previously. Therefore, we can assume the validity of Claim 6.4. $\square$

Since M_k and N_k converge uniformly toward one another, we can use arguments similar to Step 2 above. Claim 6.4 implies that the area of the boundaries of the regions Q_i is uniformly small. This is exactly what is needed to interpolate from M_k to N_k through partitioning the fundamental domain. Because

M_k and N_k project to the same minimizer in the torus, Φ descends to an r_k -sweep-out there. As a result, we have

$$\lim_{k \rightarrow \infty} (\omega(r_k) - S(r_k)) = 0.$$

We turn to the second statement of the theorem. Suppose that α is rationally dependent and that $\mathcal{M}(\alpha)$ foliates $\mathbb{R}^n$. Let M and N be two distinct elements in $\mathcal{M}(\alpha)$. Since they have maximal periodicity and α is rationally dependent, there exists $\beta \in \mathbb{Z}^n$ such that $I(\alpha, \beta) = 0$. As a result, M and N are β -invariant. Therefore, the gap bounded by M and N has an infinite volume.

Now consider a sequence of integer classes r_k converging to α . If $r_k = \alpha$ for sufficiently large k , by considering the one-parameter family φ_t of area-minimizing hypersurfaces in $\mathcal{M}(\alpha)$, we have the following:

$$S(r_k) = S(\alpha) \leq \omega(\alpha) = \omega(r_k) \leq \sup \mathbf{M}(\varphi_t) = S(\alpha).$$

It follows that all inequalities become equalities. Hence, $\omega(r_k) - S(r_k) = 0$ for sufficiently large k . Therefore, we can assume that r_k are all different from α . Let $M_k, N_k \in \mathcal{M}_{r_k}^{per}$ be two consecutive lifts of a homologically area-minimizing hypersurface in r_k as before. By translating if necessary, we can assume that they intersect a fixed bounded set. Compactness theorem 4.7 then gives

$$M_k, N_k \rightarrow M, N \in \mathcal{M}_\alpha.$$

Since $\mathcal{M}(\alpha)$ is a foliation, by the structure theorem, we must have $\mathcal{M}_\alpha = \mathcal{M}(\alpha)$. It follows that M and N belong to $\mathcal{M}(\alpha)$. As a result, either M and N are the same, or they bound a region of infinite volume.

We claim that the second alternative is impossible. Assume the contrary. Let Ω be the region of infinite volume between M and N . We may therefore choose a large compact set K such that the volume of the region between M and N within K , i.e., $\Omega \cap K$, is larger than the volume of a fundamental domain. Since M_k and N_k converge to M and N uniformly inside K , for sufficiently large k , a fundamental domain H_k bounded by M_k and N_k will almost cover the whole region $\Omega \cap K$. Consequently, we have

$$\text{Vol}(H_k) > \text{Vol}(\Omega \cap K).$$

This gives a contradiction because $\text{Vol}(H_k) = \text{Vol}(\mathbb{T}^n, g)$ is a fixed constant.

Therefore, the two sequences $\{M_k\}_k$ and $\{N_k\}_k$ converge to the same area-minimizing limit in $\mathcal{M}_\alpha$. Hence, the same argument as in Step 2 of the previous case applies. This completes the proof. $\square$

Remark 6.5. It follows from this result that $\Delta W_\alpha > 0$ is an obstruction to the existence of a minimal foliation in $\mathcal{M}_\alpha$. A more direct geometric obstruction to minimal foliation on $\mathbb{T}^n$ is the existence of a closed, contractible minimal hypersurface. This follows from the maximum principle and the fact that the lift of a closed contractible hypersurface to $\mathbb{R}^n$ is closed. A related inverse question on $\mathbb{T}^2$ was analyzed in [17], where the authors proved the existence of foliations by minimizing geodesics in any direction under a stronger assumption

of zero topological entropy. The absence of a contractible geodesic under zero topological entropy was proved in [15].

7. SUFFICIENT CONDITION FOR MINIMAL FOLIATION

In this section, we show that the convergence of ΔW_{r_k} to 0 implies the existence of a foliation by elements of $\mathcal{M}_\alpha$. This is the implication (2) $\implies$ (3) in Theorem 1.3. For convenience, we restate it here.

Theorem 7.1. *For a homological direction $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R}) \cong \mathbb{R}^n$ and a sequence of integer classes r_k converging to α , if $\Delta W_{r_k} = \omega(r_k) - S(r_k)$ converges to 0, then there exists a foliation of $\mathbb{R}^n$ by area-minimizing hypersurfaces with homological direction α .*

Proof. Since $\omega(r_k) - S(r_k) \rightarrow 0$, by choosing an appropriate subsequence and re-indexing, we can assume that $\omega(r_k) - S(r_k) \leq \frac{1}{k}$ for all k . For each k , consider a sweep-out $\Psi_k(t), t \in S^1$ of $\mathbb{T}^n$ by hypersurfaces in the class r_k such that

$$0 \leq \mathbf{M}(\Psi_k(t)) - S(r_k) \leq \frac{2}{k}. \quad (7.1)$$

In fact, it follows from the work of Chambers and Liokumovich (Theorem 1.4 in [14]) that we can assume that all these sweep-outs consist of level sets of Morse functions into S^1 . In particular, all but finitely many of the leaves are embedded hypersurfaces. Moreover, any two different leaves in the same sweep-out are disjoint.

We can lift each of these sweep-outs to the universal cover to get a foliation of $\mathbb{R}^n$ by complete, non-self-intersecting hypersurfaces. We now show that this sequence of sweep-outs converges, after passing to a subsequence, to a foliation of $\mathbb{R}^n$ by area-minimizing hypersurfaces.

For a small open ball $U = B(x, r) \subset \mathbb{T}^n$, consider its pre-image $\bar{U}$ inside the unit cube $[0, 1]^n$ under the projection map $\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n$. For any k , since Ψ_k is a sweep-out, there must exist a $t_k \in S^1$ such that the slice $\Psi_k(t_k)$ divides $\bar{U}$ into two regions of equal volume. Inequality (7.1) together with the periodicity of the metric on $\mathbb{R}^n$ implies that for any compact set $K \subset \mathbb{R}^n$, there exists a constant $C_K > 0$ such that

$$\mathbf{M}(\Psi_k(t_k) \llcorner K) < C_K$$

for all k . The Federer-Fleming compactness theorem for integral currents of bounded mass then implies that a subsequence (still denoted by $\Psi_k(t_k)$) converges in the flat topology on compact sets to $\Psi \in \mathcal{R}_{n-1}^{loc}(\mathbb{R}^n)$. We first establish the following three claims about Ψ :

Claim 7.2. $\Psi \in \mathcal{R}_{n-1}^{loc}(\mathbb{R}^n)$ is area-minimizing.

Proof of claim. If this is not the case, then there exists a ball B such that Ψ is not area-minimizing inside B . Let X be an integral current in B such that

$\partial X = \partial(\Psi \cap B)$ and:

$$\mathbf{M}(\Psi \llcorner B) - \mathbf{M}(X) = \delta > 0.$$

Since $\Psi_k(t_k) \rightarrow \Psi$ in the flat topology, by the lower-semicontinuity of the mass with respect to the flat convergence, we have:

$$\mathbf{M}(\Psi \llcorner B) \leq \mathbf{M}(\Psi_k(t_k) \llcorner B) + \frac{\delta}{2}$$

for sufficiently large k . Let us write

$$\Psi \llcorner B = \Psi_k(t_k) \llcorner B + S + \partial(T).$$

It follows that X decomposes as

$$X = \Psi_k(t_k) \llcorner B + S + \partial T'.$$

Because of the flat convergence in B , for large k we have $\mathbf{M}(S) < \frac{\delta}{4}$. Consider the following current:

$$\tilde{\Psi}_k = \Psi_k(t_k) + \partial T' \in r_k.$$

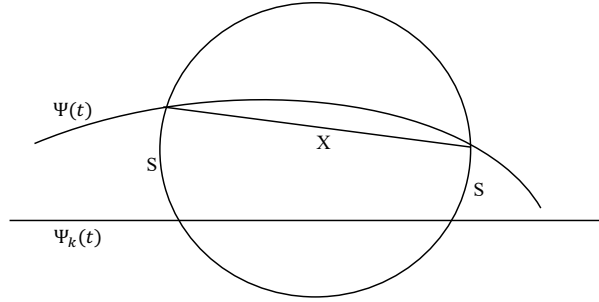


FIGURE 5. The local replacement of $\Psi_k(t_k)$ by X inside the ball B decreases the mass.

We have:

$$\begin{aligned} S(r_k) &\leq \mathbf{M}(\tilde{\Psi}_k) \leq \mathbf{M}(\Psi_k(t_k) \llcorner (\mathbb{R}^n \setminus B)) + \mathbf{M}(S) + \mathbf{M}(X) \\ &= \mathbf{M}(\Psi_k(t_k) \llcorner (\mathbb{R}^n \setminus B)) + \mathbf{M}(\Psi \llcorner B) - \delta + \mathbf{M}(S) \\ &\leq \mathbf{M}(\Psi_k(t_k) \llcorner (\mathbb{R}^n \setminus B)) + \mathbf{M}(\Psi_k(t_k) \llcorner B) + \frac{\delta}{2} - \delta + \mathbf{M}(S) \\ &\leq \mathbf{M}(\Psi_k(t_k)) + \frac{\delta}{2} - \delta + \frac{\delta}{4} \\ &\leq S(r_k) + \frac{2}{k} - \frac{\delta}{4}. \end{aligned} \tag{7.2}$$

The last inequality above follows from (7.1). This is impossible when k is sufficiently large. Therefore, Ψ is area-minimizing. It follows from the regularity theory for area-minimizing hypersurfaces that Ψ is smooth. $\square$

Claim 7.3. Ψ passes through $\bar{U}$.

Proof of claim. This is a simple consequence of the isoperimetric inequality. For a fixed geodesic ball $U = B(x, r)$, there is a positive constant δ such that for all smooth hypersurfaces Σ dividing U into two pieces of equal volume, we have $\mathbf{M}(\Sigma \llcorner U) \geq \delta$. Since all $\Psi_k(t_k)$ bisect U , so does the flat limit Ψ , hence

$$\mathbf{M}(\Psi \llcorner U) \geq \delta > 0.$$

This shows that Ψ intersects $\bar{U}$. $\square$

Claim 7.4. Ψ satisfies the Birkhoff property.

Proof of claim. This follows from the fact that Ψ arises as the limit of embedded hypersurfaces. Suppose, for contradiction, that for some integer translation τ_η , we have that

$$\Psi \cap \tau_\eta \Psi \neq \emptyset \quad \text{and} \quad \Psi \neq \tau_\eta \Psi.$$

Moreover, this intersection is transverse, otherwise $\Psi = \tau_\eta \Psi$. Since $\Psi = \lim_{k \rightarrow \infty} \Psi_k(t_k)$, it follows that for sufficiently large k :

$$\Psi_k(t_k) \cap \tau_\eta(\Psi_k(t_k)) \neq \emptyset \quad \text{and} \quad \Psi_k(t_k) \neq \tau_\eta(\Psi_k(t_k)).$$

However, this contradicts the fact that the projection of $\Psi_k(t_k)$ onto the torus is embedded and therefore has no self-intersection. $\square$

From the three claims above, we deduce that $\Psi \in \mathcal{M}$. By the continuity of homological vectors with respect to the flat convergence, it follows further that $\Psi \in \mathcal{M}_\alpha$.

We next show that for any finite collection of open sets $U_1, U_2, \dots, U_k$ contained inside a fixed fundamental domain, there exist pairwise disjoint minimizers

$$\Sigma_1, \Sigma_2, \dots, \Sigma_k \in \mathcal{M}_\alpha$$

such that $\Sigma_j \cap U_j \neq \emptyset$ for every j . Choose pairwise disjoint open balls

$$B_j \subset U_j, 1 \leq j \leq k.$$

By the three claims above, there exists a sequence of sweep-outs Ψ_i by level sets of Morse functions, together with parameters t_i , such that $\Psi_i(t_i) \rightarrow \Sigma_1$ and $\Sigma_1 \cap B_1 \neq \emptyset$. Passing to subsequences if needed, we may repeat the argument to obtain a sequence of leaves converging to Σ_2 that intersects B_2 , while ensuring that Σ_1 and Σ_2 are disjoint. Iterating this process, and each time replacing a full sequence of sweep-outs by a suitable subsequence, we obtain $\Sigma_1, \dots, \Sigma_k$ as claimed.

Now let $\{y_i\}_{i=1}^\infty$ be a sequence of points dense in a fixed fundamental domain. By the claim above, for every k , there exist pairwise disjoint minimizers

$$\Sigma_1^k, \Sigma_2^k, \dots, \Sigma_k^k$$

such that

$$\Sigma_j^k \cap B\left(y_j, \frac{1}{k}\right) \neq \emptyset, \forall 1 \leq j \leq k.$$

Passing to limits, we may assume that $\Sigma_j^k \rightarrow \Sigma_j \in \mathcal{M}_\alpha$. It follows that $y_j \in \Sigma_j$ for all $j \in \mathbb{N}$. Furthermore, suppose that for some $i \neq j$:

$$\Sigma_i \neq \Sigma_j \quad \text{and} \quad \Sigma_i \cap \Sigma_j \neq \emptyset,$$

then for sufficiently large k , we would have

$$\Sigma_i^k \neq \Sigma_j^k \quad \text{and} \quad \Sigma_i^k \cap \Sigma_j^k \neq \emptyset,$$

contradicting the construction of Σ_i^k and Σ_j^k . Therefore, any two such limit minimizers either coincide or are disjoint.

We have thus obtained a collection of disjoint minimizers in $\mathcal{M}_\alpha$ whose union is dense in a fundamental domain. Taking the closure of this collection yields a foliation by elements of $\mathcal{M}_\alpha$. This completes the proof. $\square$

Remark 7.5. One may ask whether the convergence of ΔW_{r_k} to 0 implies the existence of a foliation by elements of $\mathcal{M}(\alpha)$. We can view it as the converse of Theorem 1.6, and it is stronger than the conclusion of Theorem 7.1. However, this implication is false, as the following example shows.

Example 7.6. Consider $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ equipped with the so-called *Liouville metric* of the form

$$ds^2 = f(y)(dx^2 + dy^2)$$

where

$$f : \mathbb{R} \rightarrow \mathbb{R}_{>0}, \quad f(x+1) = f(x)$$

is a smooth, strictly positive 1-periodic function. This metric is invariant under any real horizontal translation $\varphi_t(\vec{x}) = \vec{x} + t\vec{e}_1$. Therefore, for every direction $r \in H_1(\mathbb{T}^2, \mathbb{Z})$ that is not parallel to the y -axis, we can take a homologically area-minimizing geodesic in r , lift it to $\mathbb{R}^2$ and translate it by φ_t . This produces a foliation, and hence $\Delta W_r = 0$ for all such r . Now consider the remaining direction $\alpha = [0, 1]$. If $f(y)$ is not constant, the following computation shows that $\mathcal{M}(\alpha)$ consists of only finitely many affine horizontal lines of the form $y = y_0$ for some y_0 at which f attains its global minimum. Let

$$\gamma(t) = (x(t), y(t)), \quad t \in \mathbb{R}$$

be a unit-speed parametrization of a geodesic. A standard computation shows that the geodesic equation:

$$\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0,$$

in the global coordinates becomes:

$$\ddot{x} + \frac{f'(y)}{f(y)} \dot{x} \dot{y} = 0$$

and

$$\ddot{y} - \frac{f'(y)}{2f(y)} \dot{x}^2 + \frac{f'(y)}{2f(y)} \dot{y}^2 = 0.$$

From these equations, one can check that every horizontal line passing through the critical points of $f(y)$ on the y -axis is a geodesic, and the distance-minimizing ones are exactly at the global minimizers of $f(y)$. To see that there is no other minimizer, note that for any minimizing geodesic γ , the real horizontal translation $\tau_t\gamma$ is also minimizing for all t . Thus γ and $\tau_t\gamma$ must either coincide or be disjoint for each t . For a $[0, 1]$ -periodic geodesic, this only happens when γ is a horizontal line.

These minimizers are φ_t invariant. In this case, $\mathcal{M}(\alpha)$ does not form a foliation, even though $\Delta W_{r_k} = 0$ for any r_k not parallel to α .

Indeed, if we look at the heteroclinic minimizers in $\mathcal{M}_\alpha$, they will form a foliation: take any heteroclinic minimizer between two consecutive minimizers in $\mathcal{M}(\alpha)$, it is not horizontal. We can then use φ_t to translate the heteroclinic and foliate the whole space.

This picture can be generalized to any dimension.

8. PROOF OF THEOREM 1.5

We want to construct a metric g , a rationally dependent class α , and a sequence of integer classes $r_k \rightarrow \alpha$ such that:

- (1) There exists a foliation of $\mathbb{T}^n$ by elements in $\mathcal{M}_\alpha$.
- (2) The sequence $\Delta W_{r_k} = \omega(r_k) - S(r_k)$ is bounded away from zero.

Roughly speaking, we want a foliation by elements of $\mathcal{M}_\alpha$ containing at least one leaf that cannot be approximated by any sequence of closed minimizers in r_k .

We first need the following construction of Bangert (cf. [11], Theorem 3.1) of a metric where no minimal foliation exists.

Proposition 8.1. *Suppose that g_{flat} is the Euclidean metric on $\mathbb{R}^n$ and $V \subset \mathbb{R}^n$ is a $\mathbb{Z}^n$ -periodic, non-empty open set. There exists a Riemannian metric g that coincides with g_{flat} outside V such that $\mathbb{R}^n$ does not admit any C^0 foliation by minimal hypersurfaces.*

Proof. (Sketch). The idea is to glue g_{flat} to a metric that becomes very large on a region V while agreeing with g_{flat} elsewhere.

Let $B(x, 2r) \subset V$ be an open Euclidean ball. Choose a smooth cut-off function

$$\varphi : \mathbb{R}^n \rightarrow [0, 1]$$

with support contained in $B(x, 2r)$ and satisfying $\varphi|_{B(x, r)} = 1$. Here, all the balls are with respect to the flat Euclidean metric.

Define the glued metric

$$g_\lambda = (1 - \varphi)g_{flat} + \lambda^2\varphi g_{flat}.$$

We claim that for λ sufficiently large, there is no foliation of $\mathbb{R}^n$ by g_λ -minimal hypersurfaces. Assume the contrary, and let $\mathcal{F}$ be such a minimal foliation.

Using the localized Lipschitz normals to the leaves (cf. [44]), one sees that every leaf Σ of $\mathcal{F}$ is area-minimizing with respect to g_λ .

Since $\mathcal{F}$ is a foliation, there is one leaf, say Σ , passing through x . By construction, the metric g_λ is equal to $\lambda^2 g_{flat}$ inside $B(x, r)$. It follows from the monotonicity formula that

$$\text{Area}(\Sigma \cap B(x, r)) \geq (\lambda r)^{n-1} \omega_{n-1},$$

where ω_{n-1} is the volume of the unit ball in $\mathbb{R}^{n-1}$.

Let $S(x, 3r) = \partial B(x, 3r)$ be the boundary of the ball, which is divided into two parts S_1 and S_2 with the same boundary as $\Sigma \cap B(x, 3r)$. As Σ is area-minimizing, by using S_1 and S_2 as competitors, we have

$$\begin{aligned} \text{Area}(\Sigma \cap B(x, 3r)) &\leq \min\{\text{Area}(S_1), \text{Area}(S_2)\} \\ &\leq \frac{1}{2} \text{Area}(S(x, 3r)) \leq \frac{3^{n-1} r^{n-1}}{2} \omega_{n-1}. \end{aligned} \quad (8.1)$$

As a result:

$$\lambda^{n-1} r^{n-1} \omega_{n-1} \leq \text{Area}(\Sigma \cap B(x, r)) \leq \text{Area}(\Sigma \cap B(x, 3r)) \leq \frac{3^{n-1} r^{n-1}}{2} \omega_{n-1}.$$

By choosing λ sufficiently large, we get a contradiction. As a result, no such g_λ -minimal foliation can exist. $\square$

We can now finish the proof of Theorem 1.5: We will first construct such a metric as a product $g = h \times d\theta^2$ on $\mathbb{T}^3 = \mathbb{T}^2 \times S^1$. The construction can be extended to higher dimensions. According to Proposition 8.1, we can find a metric h on the two-torus $\mathbb{T}^2$ such that no minimal foliation by geodesics exists.

Consider the universal cover $\mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R}$ and fix a rationally dependent class of the form $\alpha = [\alpha_1, \alpha_2, 0]$. For the torus, the homology intersection pairing

$$I : H_1(\mathbb{T}^3, \mathbb{R}) \times H_2(\mathbb{T}^3, \mathbb{R}) \rightarrow \mathbb{R}$$

is induced from the dot product. This means that for $\beta = [0, 0, 1] \in H_1(\mathbb{T}^3, \mathbb{Z})$ we have $I(\alpha, \beta) = 0$. Therefore, for any integral current $T \in \alpha$, we have $\tau_\beta \tilde{T} = \tilde{T}$ for any lift $\tilde{T}$ of T .

Since the lift $\tilde{g}$ of g to $\mathbb{R}^3$ is the product of the lift of h and the flat metric $d\theta^2$ on S^1 , we see that $\tilde{g}$ is translationally invariant in the β direction:

$$\varphi_t^*(\tilde{g}) = \tilde{g}, \quad \forall t \in \mathbb{R}$$

where $\varphi_t(x) = x + t\beta$.

We first show that there is a foliation of $\mathbb{R}^3$ by the elements of

$$\mathcal{M}(\alpha) \cup \mathcal{M}(\alpha, \beta).$$

Since the metric is flat in the vertical direction, any minimizer $T \in \mathcal{M}(\alpha)$ must be of the form $\gamma \times \mathbb{R}$ where γ is a minimizer in $\mathbb{R}^2$ with respect to the direction $[\alpha_1, \alpha_2]$. Since h is chosen so that there is no minimal foliation on $\mathbb{R}^2$, we see that $\mathcal{M}(\alpha)$ is not a foliation of $\mathbb{R}^3$.

Consider a gap G bounded by neighboring periodic $\Sigma^-, \Sigma^+ \in \mathcal{M}(\alpha)$. By the structure theorem, Theorem 4.10, we can find a heteroclinic minimizer

$$\Sigma \in \mathcal{M}(\alpha, \beta)$$

approaching Σ^- in the negative β direction and Σ^+ in the positive β direction. Since the metric is vertically invariant, $\tau_t \Sigma$ is area-minimizing for all $t \in \mathbb{R}$. Furthermore, by the characterization of minimizers in $\mathcal{M}(\alpha, \beta)$, all these translations are disjoint. As a result, they foliate the gap G . Repeating this for every gap, we get a foliation of the entire space.

Next, consider a sequence of integer classes $r_k = [a_k, b_k, 0]$ converging to α . We claim that the sequence:

$$\Delta W_{r_k} = \omega(r_k) - S(r_k)$$

does not converge to 0. Assume the contrary that

$$\lim_{r_k \rightarrow \alpha} \Delta W_{r_k} = 0.$$

We then proceed as in the proof of Theorem 1.3 to obtain a foliation of $\mathbb{R}^3$ by area-minimizing hypersurfaces in $\mathcal{M}_\alpha$. In fact, from the proof, any leaf Ψ of this foliation is a limit of leaves $\Psi_k(t_k)$ of sweep-outs in r_k . Since

$$I(r_k, \beta) = r_k \cdot \beta = 0$$

for all k , Lemma 4.6 applied to the closed normal current $\Psi_k(t_k) \in r_k$ shows that its lift is invariant under τ_β . Consequently, the limit Ψ is also τ_β -invariant. This means that the limiting foliation is formed by elements in $\mathcal{M}(\alpha)$. This contradicts the previous paragraph.

Therefore, the sequence ΔW_{r_k} is uniformly bounded from below away from 0. A similar construction extends to any $\mathbb{T}^n$ for $n > 3$. This completes the proof of Theorem 1.5.

9. MINIMAL HYPERSURFACE INSIDE THE GAP

In this section, we prove Theorem 1.7 and Theorem 1.8. Our goal is to construct a new minimal hypersurface lying inside the gap formed by two consecutive elements in $\mathcal{M}_\alpha$, when α is totally irrational. Assume that two such consecutive minimizers Σ_0 and Σ_1 in $\mathcal{M}_\alpha$ bound a gap G . Without loss of generality, assume further that $0 \in G$. We now work under the following assumption:

There is no closed minimal hypersurface inside the gap G . (★)

This assumption reduces the number of cases in the min-max argument. At the end of the section, we will remove this assumption.

From the analysis in Section 6, we know that G projects injectively to $\mathbb{T}^n$ under the covering map and therefore has finite volume. Consequently, Σ_0 and Σ_1 are asymptotic outside a sufficiently large compact ball.

The idea of the proof of Theorem 1.7 is as follows. We will first construct a compact minimal hypersurface Γ_R , possibly with boundary contained in $\partial B_R(0)$, that lies between

$$\Sigma_0^R = \Sigma_0 \cap B_R \text{ and } \Sigma_1^R = \Sigma_1 \cap B_R.$$

We then consider the limit of the sequence of hypersurfaces Γ_R as $R \rightarrow \infty$. A subtle difficulty arises from the possibility that the hypersurfaces Γ_R may collapse to one of the minimizers, either Σ_0 or Σ_1 .

To exclude this possibility, we obtain a uniform area bound on a fixed compact set. This is done in the following proposition:

Proposition 9.1. *There exists a positive constant $C > 0$ and a sufficiently large radius R_0 such that the following holds. For all $R > R_0$ and for every one-parameter family $\{\Sigma_t^R\}_{t=0}^1$ of generalized hypersurfaces with boundary contained in ∂B_R , connecting Σ_0^R and Σ_1^R , we have the following:*

$$\sup_{t \in [0,1]} \mathbf{M}(\Sigma_t^R) - \max\{\mathbf{M}(\Sigma_0^R), \mathbf{M}(\Sigma_1^R)\} > C. \quad (9.1)$$

Proof. We argue by contradiction. Suppose that no such constants exist. Then we can find a sequence of radii $R_k \rightarrow \infty$ and a sequence of one-parameter families of hypersurfaces $\{\Sigma_t^{R_k}\}_k$ such that, after passing to a subsequence if necessary, the following holds:

$$\sup_{t \in [0,1]} \mathbf{M}(\Sigma_t^{R_k}) - \max\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} < \frac{1}{k}$$

for all $k \in \mathbb{N}$. We now show that this forces the gap G to be foliated by area-minimizing hypersurfaces. Let $U \subset G$ be a small open ball. For all large k , we have $U \subset G \cap B_{R_k}$. Since $\Sigma_t^{R_k}$ is a sweep-out, there exists $t_k \in [0, 1]$ such that $\Sigma_{t_k}^{R_k}$ divides U into two regions of equal volume. It follows from the isoperimetric inequality that any local limit of the sequence $\Sigma_{t_k}^{R_k}$ must intersect U . Similar arguments as in the proof of Theorem 7.1 establish that $\Sigma_{t_k}^{R_k}$ converges on compact sets to an area-minimizing hypersurface Σ inside G that intersects U .

Since α is totally irrational, G projects injectively onto the torus. Equivalently, for any integer translation τ_η with $\eta \neq 0$, we have $\tau_\eta(G) \cap G = \emptyset$. As a consequence, for any minimizing hypersurface Σ inside G , we also have $\tau_\eta \Sigma \cap \Sigma = \emptyset$. In particular, $\Sigma \in \mathcal{M}_\alpha$. By repeating this argument for any open set $U \subset G$ and using the fact that $\mathcal{M}_\alpha$ is a lamination when α is totally irrational, we obtain a foliation of G by elements of $\mathcal{M}_\alpha$.

This contradicts the assumption that G is a gap in the lamination in $\mathcal{M}_\alpha$ and completes the proof. $\square$

Remark 9.2. In fact, it follows from the same interpolation technique in Section 6 that the constant C can be chosen to be $C = \Delta W_\alpha$. Indeed, if there

exists a sequence of radii and a sequence of corresponding sweep-outs that satisfy

$$\sup_{t \in [0,1]} \mathbf{M}(\Sigma_t^{R_k}) - \max\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} < C_0 < \Delta W_\alpha,$$

then we can modify to construct sweep-outs in a sequence of homology classes $r_k \rightarrow \alpha$ with $\liminf \Delta W_{r_k} < \Delta W_\alpha$, contradicting the definition of the Mather energy barrier of α .

Let σ_0^R and σ_1^R be the boundaries of Σ_0^R and Σ_1^R .

Proposition 9.3. *For any $\epsilon > 0$ sufficiently small, there exists $R_\epsilon > 0$ such that for all $R > R_\epsilon$, there exists a $(n-2)$ -dimensional integral current σ^R with support contained in $\partial B_R \cap G$ that satisfies the following properties:*

- (1) *There are at least two smooth, area-minimizing hypersurfaces Γ_0^R and Γ_1^R with the same boundary σ^R .*
- (2) *Γ_0^R and Γ_1^R meet only along their boundary, and hence bound an open domain.*
- (3) *The following inequality holds:*

$$\min\{\mathbf{M}(\Sigma_0^R), \mathbf{M}(\Sigma_1^R)\} - \epsilon < \mathbf{M}(\Gamma_0^R), \mathbf{M}(\Gamma_1^R) < \max\{\mathbf{M}(\Sigma_0^R), \mathbf{M}(\Sigma_1^R)\} + \epsilon.$$

- (4) *For any one-parameter family $\{\phi_t\}_{t \in [0,1]}$ of hypersurfaces with $\partial\phi_t = \sigma^R$ connecting Γ_0^R and Γ_1^R , it holds that:*

$$\sup_{t \in [0,1]} \mathbf{M}(\phi_t) \geq \max\{\mathbf{M}(\Sigma_0^R), \mathbf{M}(\Sigma_1^R)\} + C, \quad (9.2)$$

where C is the same constant found in Proposition 9.1.

Proof. By Lemma 6.3, we can choose R_ϵ such that $\mathbf{M}(\partial B_R \cap G) < \epsilon$ for all $R > R_\epsilon$. Assume for contradiction that there exists a sequence of radii $\{R_k\}_k$ for which no such σ^{R_k} exists.

For any k , since $\sigma_0^{R_k}$ and $\sigma_1^{R_k}$ bound an open set on ∂B_{R_k} and $\mathcal{F}(\sigma_0^{R_k}, \sigma_1^{R_k}) < \epsilon$, we can find a one-parameter family of smooth boundaries $\{\sigma_t^{R_k}\}_{t=0}^1$ connecting them in such a way that:

$$\mathcal{F}(\sigma_t^{R_k}, \sigma_s^{R_k}) < \epsilon \quad (9.3)$$

for all $s, t \in [0, 1]$. For each t , by solving the Plateau problem with boundary $\sigma_t^{R_k}$, we obtain at least one area-minimizing hypersurface $\Gamma_t^{R_k}$. It follows from (9.3) and an area comparison argument that:

$$\min\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} - \epsilon < \mathbf{M}(\Gamma_t^{R_k}) < \max\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} + \epsilon.$$

Therefore, assuming the contradiction, $\Gamma_t^{R_k}$ is unique for every t . As a result, by compactness for area-minimizing currents, the family $\{\Gamma_t^{R_k}\}_t$ gives a continuous sweep-out of hypersurfaces connecting $\Sigma_0^{R_k}$ and $\Sigma_1^{R_k}$ with mass bounded from above by

$$\max\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} + \epsilon.$$

This contradicts Proposition 9.1 when $\epsilon < C$.

As a result, for sufficiently large R , there exists σ^R , together with two distinct Plateau solutions Γ_0^R and Γ_1^R with boundary σ^R . The standard cut-and-paste argument for area-minimizing hypersurfaces shows that they only intersect along their boundary.

If the conclusion (4) does not hold for any such σ^R when R is sufficiently large, we can then construct a one-parameter family of hypersurfaces from Σ_0^R to Σ_1^R that violates (9.1) as follows. First, choose a flat-continuous one-parameter family of boundaries $\{\sigma_t^R\}_t$ on $\partial B_R \cap G$ that connects σ_0^R and σ_1^R , then find an area-minimizing hypersurface for each of them. By area comparison, for sufficiently large R , any such hypersurface Γ_t^R must have mass bounded from above by

$$\max\{\mathbf{M}(\Sigma_0^R), \mathbf{M}(\Sigma_1^R)\} + C.$$

For those σ_t^R that bound multiple area-minimizing solutions, by the assumption, we can interpolate from one solution to another by a boundary-constrained family of integral currents of mass also bounded by the same constant. Combining these facts, we get a contradiction. This proves the proposition. $\square$

We can now finish the proof of Theorem 1.7 under the assumption $(\star)$ as follows.

Proof. By Proposition 9.3, we can find a sequence of radii $R_k \rightarrow \infty$ and a sequence of boundaries $\{\sigma^{R_k}\}_k$ contained in ∂B_{R_k} that satisfies (1), (2), (3) and (4) above. We have that $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$ stay inside the gap between $\Sigma_0^{R_k}$ and $\Sigma_1^{R_k}$. Since they are stable, we can perturb them in an arbitrarily small neighborhood to create a mean-convex open ball U_k , with $\sigma^{R_k} \subset \partial U_k$ and $\Gamma_0^{R_k}, \Gamma_1^{R_k} \subset \text{Int}(U_k)$. It follows from (9.2) that

$$W_{\sigma^{R_k}}(\Gamma_0^{R_k}, \Gamma_1^{R_k}) \geq \max\{\mathbf{M}(\Sigma_0^{R_k}), \mathbf{M}(\Sigma_1^{R_k})\} + C.$$

We can now apply Montezuma's min-max theorem, Theorem 2.7, to produce a hypersurface Γ^{R_k} with boundary σ^{R_k} and of mass equal to $W_{\sigma^{R_k}}(\Gamma_0^{R_k}, \Gamma_1^{R_k})$. Note that to obtain the boundary regularity of the min-max varifold along σ^{R_k} , we need that σ^{R_k} is contained in the boundary of a region with a strictly convex boundary. In fact, the argument only requires local convexity of the region near σ^{R_k} . In our setting, this can be guaranteed by a small perturbation of ∂B_{R_k} , because the metric is periodic (hence bounded) and $\mathbf{M}(\partial B_{R_k} \cap G) \rightarrow 0$.

Furthermore, under the assumption that G contains no closed minimal hypersurfaces, we must have that Γ^{R_k} consists of only one connected component with boundary σ^{R_k} . It also follows from Theorem 2.7 that Γ^{R_k} has multiplicity one.

We now prove the following claim:

Claim 9.4. *There exists R' such that for all $R_k > R'$, we have*

$$\mathbf{M}(\Gamma^{R_k} \llcorner B_{R'}) \geq \max\{\mathbf{M}(\Sigma_0^{R'}), \mathbf{M}(\Sigma_1^{R'})\} + C. \quad (9.4)$$

Proof of claim. Since the Plateau solutions $\Gamma_0^{R_k}, \Gamma_1^{R_k}$ lie inside the gap G , for any ϵ , there is R' such that for sufficiently large R_k :

$$\mathcal{F}(\Gamma_0^{R_k} \setminus B_{R'}, \Gamma_1^{R_k} \setminus B_{R'}) < \epsilon$$

and

$$\mathbf{M}(\partial B_{R'} \cap G) < \epsilon.$$

For any sweep-out $\{\Gamma_t^{R_k}\}_t$ from $\Gamma_0^{R_k}$ to $\Gamma_1^{R_k}$, we can modify the slices outside of $B_{R'}$, then reconnect along $\partial B_{R'} \cap G$ to obtain a new sweep-out whose main contribution to the width comes from the mass of the slices inside $B_{R'}$.

Since $\partial B_{R'} \cap G$ has small mass, we can modify a sweep-out from $\Gamma_0^{R_k} \llcorner B_{R'}$ to $\Gamma_1^{R_k} \llcorner B_{R'}$ to a sweep-out from $\Gamma_0^{R'}$ to $\Gamma_1^{R'}$. Therefore, combining this with Proposition 9.1, we establish (9.4) for sufficiently small ϵ . $\square$

We thus arrive at a sequence of minimal hypersurfaces $\{\Gamma^{R_k}\}_k$ with boundaries $\{\sigma^{R_k}\}_k$, whose Morse indices are bounded by one. Because the metric is periodic and therefore uniformly bounded, the sequence $\{\Gamma^{R_k}\}_k$ has uniformly bounded mass on every bounded compact set. Therefore, by Sharp's compactness theorem [42], Γ^{R_k} converges in the varifold sense on compact sets to a smooth minimal hypersurface Γ lying between Σ_0 and Σ_1 . In fact, the convergence is smooth away from at most one point. Since there is no loss of mass in the varifold convergence, the bound (9.4) is preserved in the limit, and we have:

$$\mathbf{M}(\Gamma \llcorner B_{R'}) \geq \max\{\mathbf{M}(\Sigma_0^{R'}), \mathbf{M}(\Sigma_1^{R'})\} + C. \quad (9.5)$$

This implies that Γ is not area-minimizing and hence distinct from Σ_0 and Σ_1 . This is the desired hypersurface inside G .

We now show that Γ satisfies the estimate in the first statement of Theorem 1.8 as follows. For any $\epsilon > 0$, as above we can choose R' such that (9.5) holds and that

$$\mathcal{F}(\Gamma \cap \partial B_{R'}, \Sigma_0 \cap \partial B_{R'}) < \epsilon.$$

By Lemma 6.3, we have for all sufficiently large R

$$\mathcal{F}(\Gamma \cap \partial B_R, \Sigma_0 \cap \partial B_R) < \epsilon.$$

Since Σ_0 is area-minimizing, area comparison gives:

$$\mathbf{M}(\Gamma \llcorner (B_R - B_{R'})) \geq \mathbf{M}(\Sigma_0 \llcorner (B_R - B_{R'})) - 2\epsilon.$$

Together with (9.5), we have

$$\mathbf{M}(\Gamma \llcorner B_R) \geq \mathbf{M}(\Sigma_0 \llcorner B_R) + C - 2\epsilon.$$

Since this holds for all sufficiently large R , take $\epsilon \rightarrow 0$ and note that by Remark 9.2, $C = \Delta W_\alpha$, we conclude that

$$\limsup_{R \rightarrow \infty} (\mathbf{M}(\Gamma \llcorner B_R) - \mathbf{M}(\Sigma_0 \llcorner B_R)) \geq \Delta W_\alpha.$$

$\square$

We now turn to the general case. From the preceding argument, if Γ^{R_k} contains no closed component for all k sufficiently large, then the conclusion already follows. We therefore assume that this fails, so that closed components do appear along a subsequence.

Consider the sequence of boundaries σ^{R_k} as before, together with area-minimizing hypersurfaces $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$ spanning σ^{R_k} . Together they bound an open subset G_{R_k} of G . We have that $G_{R_k} \rightarrow G$.

The strategy is to cut along a *maximal* collection of closed stable minimal hypersurfaces and then perform the min-max construction in the remaining region. The key point is that the number of stable minimal hypersurfaces that need to be removed is finite.

Proposition 9.5. *There exists a finite collection of pairwise disjoint open sets $U_{S_1}, \dots, U_{S_j} \subset G$ such that for every $1 \leq i \leq j$, $\partial U_{S_i} = S_i$ is a closed, connected, stable minimal hypersurface inside G , and such that when k is sufficiently large, for every closed minimal hypersurface Σ contained inside the interior of*

$$\Omega_{R_k} := G_{R_k} \setminus (\cup_{i=1}^j U_{S_i}),$$

we have

$$\mathbf{M}(\Sigma) \geq \mathbf{M}(\Gamma_0^{R_k}) + \mathbf{M}(\Gamma_1^{R_k}).$$

We postpone the proof of this lemma to Appendix E.

Let $\mathcal{S}$ be the collection $\{S_1, S_2, \dots, S_j\}$ of stable minimal hypersurfaces constructed in Proposition 9.5. Since $G_{R_k} \rightarrow G$, G_{R_k} contains all elements of $\mathcal{S}$ for sufficiently large k . For each R_k , if there exists a stable hypersurface X_{R_k} with boundary σ^{R_k} that does not lie completely on one side (above or below) with respect to every closed hypersurface in $\mathcal{S}$, then we can take the limit of X_{R_k} to produce a new minimal hypersurface inside the gap G . As a result, we can assume that every stable hypersurface inside G_{R_k} lies completely on one side with respect to $\mathcal{S}$.

By replacing $\Gamma_0^{R_k}$ by

$$\sup\{X \subset G_{R_k} : \partial X = \sigma^{R_k}, X \text{ is stable, } X \text{ lies below } \mathcal{S}\}$$

and $\Gamma_1^{R_k}$ by

$$\inf\{X \subset G_{R_k} : \partial X = \sigma^{R_k}, X \text{ is stable, } X \text{ lies above } \mathcal{S}\},$$

we can assume that there is no stable minimal hypersurface with boundary σ^{R_k} lying completely inside G_{R_k} .

For each k , we can apply Theorem 2.7 to

$$[\Gamma_0^{R_k}] = \left[\Gamma_1^{R_k} + \sum_{i=1}^j S_i \right] \in H_{n-1}(\Omega_{R_k}, \partial\Omega_{R_k})$$

whose representatives have the same boundary σ^{R_k} to get another minimal hypersurface Γ^{R_k} satisfying the following:

$$\mathbf{M}(\Gamma^{R_k}) > \mathbf{M}(\Gamma_1^{R_k}) + \sum_{i=1}^j \mathbf{M}(S_i). \quad (9.6)$$

Claim 9.6. *There exists a positive constant C_0 independent of k such that the following upper bound holds:*

$$\mathbf{M}(\Gamma^{R_k}) \leq \mathbf{M}(\Gamma_0^{R_k}) + \sum_{i=1}^j \mathbf{M}(S_i) + C_0.$$

Proof of claim. Since G has finite volume and Σ_0 and Σ_1 are area-minimizing, it follows from the interpolation theorem, Corollary 3.2, that for any ϵ sufficiently small, there exists R such that we can interpolate from $(\Sigma_{0\perp}(B_{R_k} \setminus B_R))$ to $\Sigma_{1\perp}(B_{R_k} \setminus B_R)$ by a sweep-out whose slices have mass at most

$$\mathbf{M}(\Sigma_{0\perp}(B_{R_k} \setminus B_R)) + \epsilon.$$

Furthermore, $\mathbf{M}(\partial B_R \cap G) \rightarrow 0$. Therefore, for every k sufficiently large, starting from a fixed sweep-out from Σ_0^R to Σ_1^R , we can change all the slices along $\partial B_R \cap G$ and then glue in a sweep-out from $\Sigma_{0\perp}(B_{R_k} \setminus B_R)$ to $\Sigma_{1\perp}(B_{R_k} \setminus B_R)$. It follows that there is a constant C_0 such that there exists a sweep-out Φ from $\Sigma_0^{R_k}$ to $\Sigma_1^{R_k}$ such that

$$\mathbf{M}(\Phi(t)) \leq \mathbf{M}(\Sigma_0^{R_k}) + C_0, \quad \forall t \in [0, 1].$$

Since $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$ converge to Σ_0 and Σ_1 , respectively, we see that a similar estimate holds for some sweep-outs between them:

$$\mathbf{M}(\Phi(t)) \leq \mathbf{M}(\Gamma_0^{R_k}) + C_0, \quad \forall t \in [0, 1]. \quad (9.7)$$

Fix one such sweep-out Φ . We can now modify Φ to obtain a sweep-out of $G_{R_k} \setminus \cup_{i=1}^j U_{S_i}$ as follows: for each $i = 1, \dots, j$, the restriction of Φ to U_{S_i} is a sweep-out of U_{S_i} by open sets V_i^t , where

$$\partial V_i^t = \Phi(t) \llcorner U_{S_i} + \text{a portion of } S_i \text{ cut by } \Phi(t).$$

As a result, we can form the following sweep-out

$$\Phi'(t) = \Phi(t) + \sum_{i=1}^j \partial V_i^t, \quad t \in [0, 1].$$

It follows that

$$\mathbf{M}(\Phi'(t)) \leq \mathbf{M}(\Phi(t)) + \sum_{i=1}^j \mathbf{M}(S_i).$$

Since the width is obtained by taking inf sup over all sweep-outs, we have

$$\mathbf{M}(\Gamma^{R_k}) \leq \sup_{t \in [0,1]} \mathbf{M}(\Phi'(t)) \leq \sup_{t \in [0,1]} \mathbf{M}(\Phi(t)) + \sum_{i=1}^j \mathbf{M}(S_i). \quad (9.8)$$

The desired inequality follows by combining (9.7) and (9.8). $\square$

Note that $\partial \Gamma^{R_k} = \sigma^{R_k}$, it follows from the min-max theorem 2.7 that we have the following decomposition:

$$\Gamma^{R_k} = W_k + \sum_{i=1}^j n_i S_i + \sum_{i=1}^m p_i \Sigma_i, \quad (9.9)$$

where $\partial W_k = \partial \Gamma^{R_k} = \sigma^{R_k}$, $\Sigma_i \subset \text{Int}(\Omega_k)$ are closed minimal hypersurfaces and $n_i, p_i \in \mathbb{Z}_{\geq 0}$. Since $\Gamma_0^{R_k}$ is a least area hypersurface with boundary σ^{R_k} , we have

$$\mathbf{M}(W_k) \geq \mathbf{M}(\Gamma_0^{R_k}).$$

It follows from Claim 9.6 that for any i such that $p_i > 0$, we have

$$\mathbf{M}(\Sigma_i) \leq \sum_{i=1}^j \mathbf{M}(S_i) + C_0.$$

Since Σ_i is a closed minimal hypersurface in the interior of Ω_{R_k} , the above upper bound violates Proposition 9.5, when k is sufficiently large. Therefore, $m = 0$ and

$$\Gamma^{R_k} = W_k + \sum_{i=1}^j n_i S_i$$

Since all S_i are stable, we have $\text{Ind}(W_k) = 1$. Under the assumption that the metric is bumpy, it follows from the proof of the multiplicity one conjecture for one-parameter min-max families in [28] (see also the introduction of [24]) that $n_i \in \{0, 1\}$ for all i .

Claim 9.7. *There exists a positive constant C independent of k such that for a subsequence of $R_k \rightarrow \infty$, we have*

$$\mathbf{M}(W_k) \geq \mathbf{M}(\Gamma_1^{R_k}) + C.$$

Proof of claim. If at least one $n_j = 0$ in (9.9), we have

$$\mathbf{M}(W_k) \geq \mathbf{M}(\Gamma_1^{R_k}) + \mathbf{M}(S_j) \geq \mathbf{M}(\Gamma_1^{R_k}) + \delta_{\mathcal{S}},$$

where $\delta_{\mathcal{S}} = \min\{\mathbf{M}(S_i) | S_i \in \mathcal{S}\}$. If this is the case for a subsequence (still denoted by R_k) of $R_k \rightarrow \infty$, we obtain a sequence of index-one minimal hypersurfaces W_k such that

$$\mathbf{M}(W_k) \geq \mathbf{M}(\Gamma_1^{R_k}) + \delta_{\mathcal{S}}.$$

Assume now that $n_j = 1$ for every j and that W_k and $\Gamma_1^{R_k}$ lie on the same side of $\mathcal{S}$ (otherwise we can take the limit of W_k again). Since W_k is unstable, there exists a small neighborhood of W_k foliated by hypersurfaces

$$W_k^t, \quad t \in [-\epsilon, \epsilon]$$

of the same boundary σ^{R_k} , such that their mean curvature vectors point away from W_k and:

$$\mathbf{M}(W_k^t) < \mathbf{M}(W_k), \quad \forall t \in [-\epsilon, \epsilon].$$

Suppose that the claim were false for all large R_k . Then we must necessarily have that

$$\mathbf{M}(W_k) < \mathbf{M}(\Gamma_0^{R_k}) + \sum_{i=1}^j \mathbf{M}(S_i).$$

Together with the strict stability of $\Gamma_0^{R_k} + \sum_{i=1}^j S_i$, we can apply Theorem 2.7 for W_k^ϵ and $\Gamma_0^{R_k} + \sum_{i=1}^j S_i$ to obtain another index-one hypersurface $\widetilde{W}_k$ disjoint from W_k (by the maximum principle) with $\partial\widetilde{W}_k = \sigma^{R_k}$.

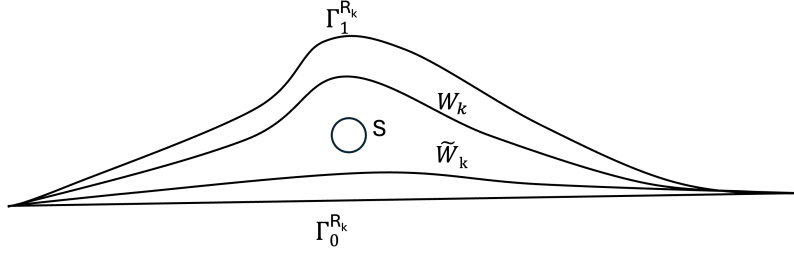


FIGURE 6. The hypersurfaces W_k and $\widetilde{W}_k$ act as barriers, forcing the existence of another stable hypersurface with boundary σ^{R_k} .

Now by using W_k and $\widetilde{W}_k$ as barriers, we can construct a stable hypersurface with the same boundary σ^{R_k} lying completely inside G_{R_k} , contradicting the choices of $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$. This completes the proof of the claim. $\square$

We can now proceed similarly as in the proof of Claim 9.4 to obtain an analogue of the inequality (9.4) with W_k in place of Γ^{R_k} : there are positive constants $C, R' > 0$ such that for all k with $R_k > R'$ we have

$$\mathbf{M}(W_k \llcorner B_{R'}) \geq \max\{\mathbf{M}(\Sigma_0^{R'}), \mathbf{M}(\Sigma_1^{R'})\} + C.$$

Note that by the construction, W_k has index at most one and uniformly bounded area on compact subsets. Therefore, Sharp's compactness theorem [42] applies to the sequence $\{W_k\}_k$, yielding a limiting hypersurface Γ that satisfies the same estimate as above. In particular, Γ is different from Σ_0 and Σ_1 and lies entirely inside G . This completes the proof of Theorem 1.7.

We are now in the position to prove the second statement of Theorem 1.8. Fix a positive constant ϵ . For each open set U_{S_i} with strictly stable boundary $S_i \in \mathcal{S}$, we can find a one-parameter sweep-out that deforms S_i to the zero current:

$$\Phi_i : [0, 1] \rightarrow \mathcal{Z}_{n-1}(U_{S_i}; \mathbf{F}, \mathbb{Z}), \quad \Phi_i(0) = S_i, \Phi_i(1) = 0$$

such that

$$W_{\partial}(U_i) + \epsilon \geq \sup_{t \in [0, 1]} \mathbf{M}(\Phi_i(t)).$$

In the set-up above, for any sweep-out Φ^{R_k} from $\Gamma_0^{R_k}$ to $\Gamma_1^{R_k} + S_1 + \dots + S_j$, we can combine with $\Phi_1, \dots, \Phi_j$ to make a sweep-out Φ^k from $\Gamma_0^{R_k}$ to $\Gamma_1^{R_k}$ of the entire region between them:

$$\Gamma_0^{R_k} \xrightarrow{\Phi^{R_k}} \Gamma_1^{R_k} + S_1 + \dots + S_j \xrightarrow{\Gamma_1^{R_k} + \Phi_1 + \dots + \Phi_j} \Gamma_1^{R_k}.$$

Since $\Gamma_1^{R_k}$ is disjoint from $\Phi_i(t)$ for all i, t , we have

$$\max\{\sup \mathbf{M}(\Phi^{R_k}(t)), \sum_{i=1}^j \sup \mathbf{M}(\Phi_i(t)) + \mathbf{M}(\Gamma_1^{R_k})\} \geq \sup \mathbf{M}(\Phi^k(t)).$$

Choose Φ^{R_k} such that

$$\mathbf{M}(\Gamma^{R_k}) + \epsilon \geq \sup \mathbf{M}(\Phi^{R_k}(t)).$$

It follows that

$$\max\{\mathbf{M}(\Gamma^{R_k}) + \epsilon, \mathbf{M}(\Gamma_1^{R_k}) + \sum_{i=1}^j W_{\partial}(U_{S_i}) + j\epsilon\} \geq \sup \mathbf{M}(\Phi^k(t)).$$

By Proposition 9.1, we have

$$\sup \mathbf{M}(\Phi^k(t)) \geq \Delta W_{\alpha} + \mathbf{M}(\Gamma_0^{R_k})$$

for sufficiently large k . Since Γ^{R_k} consists of W_k together with some of the S_i , each with multiplicity one, it follows that

$$\mathbf{M}(W_k) + \sum_{i=1}^j \mathbf{M}(S_i) \geq \mathbf{M}(\Gamma^{R_k}).$$

As a result

$$\max\{\mathbf{M}(W_k) + \sum_{i=1}^j \mathbf{M}(S_i) + \epsilon, \mathbf{M}(\Gamma_1^{R_k}) + \sum_{i=1}^j W_{\partial}(U_{S_i}) + j\epsilon\} \geq \Delta W_{\alpha} + \mathbf{M}(\Gamma_0^{R_k}).$$

By taking $k \rightarrow \infty$ and sending $\epsilon \rightarrow 0$, we see that if $\sum_{i=1}^j W_{\partial}(U_{S_i}) < \Delta W_{\alpha}$, then

$$\limsup_{R \rightarrow \infty} (\mathbf{M}(\Gamma_{\perp} B_R) - \mathbf{M}(\Sigma_0 B_R)) \geq \Delta W_{\alpha} - \sum_{i=1}^j \mathbf{M}(S_i).$$

This completes the proof of Theorem 1.8.

10. OPEN PROBLEMS

We conclude with several open questions that naturally emerge from our work.

10.1. The function ΔW_α . For a totally irrational class $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$, we conjecture that the limit

$$\Delta W_\alpha = \lim_{r_k \rightarrow \alpha} \Delta W_{r_k} = \lim_{r_k \rightarrow \alpha} (\omega(r_k) - S(r_k))$$

exists and is independent of the sequence r_k . Equivalently, the asymptotic min-max excess in rational directions approaching α should be an intrinsic quantity associated only with α rather than a particular sequence of rational classes approximating it. If this is the case, we can then extend the discrete function

$$r \rightarrow \Delta W_r, \quad r \in H_{n-1}(T^n, \mathbb{Z})$$

continuously to totally irrational classes, with the property that it vanishes if and only if there is a foliation by minimal hypersurfaces with direction α . In the case of the Hamiltonian twist map and geodesics on tori, the corresponding Mather energy barrier and Peierls's barrier were shown to be continuous at irrational frequencies, see [31] and [32]. An even stronger form of the question is whether the extended function admits a modulus of continuity depending only on the geometry of the ambient metric. The same question regarding the Peierls's barrier was analyzed in [32].

We also expect that the collection of quantities $\{\Delta W_r\}, r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ may contain global geometric information about the ambient metric. For instance, the simultaneous vanishing of barriers would imply that there exists a foliation of $\mathbb{T}^n$ by closed, homologically area-minimizing hypersurfaces in all primitive homology classes. Bangert [9] conjectured that this property implies the flatness of the Riemannian metric.

10.2. Minimal hypersurfaces inside gaps. Theorem 1.7 shows that for a generic metric, in the totally irrational setting, whenever a minimal lamination by area-minimizing hypersurfaces contains gaps, one can find complete, non-compact minimal hypersurfaces inside these gaps that are not area-minimizing. The genericity of the metric is used in the proof to ensure that the number of disjoint, stable hypersurfaces in the set S that we remove is finite. We conjecture that the same conclusion should remain valid even without the bumpiness assumption. A natural further question is to investigate the Morse index of such non-compact hypersurfaces constructed by Theorem 1.7 inside gaps. The construction gives a non-area-minimizing hypersurface Γ of index at most one, but it is not clear whether Γ must actually be unstable.

Another closely related open problem is to analyze the gap in minimal lamination in the case where α is rationally dependent. In the proof of Theorem

1.7, we rely crucially on the fact that when α is rationally independent, the corresponding gaps necessarily have finite volume and hence their boundaries converge to each other uniformly. This phenomenon is not guaranteed in the rationally dependent setting. It would be interesting if one could produce a non-minimizing hypersurface in this case. A natural way to overcome this difficulty is to work not in the universal cover $\mathbb{R}^n$, but in a suitable intermediate normal cover adapted to the rational dependencies of α , where the gap has finite volume, and then repeat the constructions carried out here.

10.3. The Birkhoff property. The main objects studied in this paper are area-minimizing hypersurfaces that satisfy Conditions (4.1) and (4.2). These conditions arise naturally from our goal of studying minimal laminations on the torus. In the non-parametric setting, for variational problems with an elliptic integrand, Bangert raised the question of whether the Birkhoff property can be forced by a weaker geometric assumption on the minimizer. In particular, he proved that minimizers lying entirely between two parallel affine hyperplanes with a totally irrational normal vector are Birkhoff. It would be interesting to investigate whether a similar phenomenon occurs in the parametric setting of minimal hypersurfaces.

10.4. Generalization to other manifolds. We expect that many of the arguments developed in this paper can be carried over to a broader class of manifolds with abelian fundamental groups, for example, on the product $\mathbb{T}^n \times S^m$.

There are also related results in the non-abelian setting. It is shown in [12] that when the fundamental group is *residually finite* and there exists a function (called the cocycle function) $\varphi : \pi_1(M) \rightarrow \mathbb{R}$ such that

$$\varphi(gg') = \varphi(g) + \varphi(g'),$$

one can still prove the existence of minimal laminations in any direction governed by the corresponding level set of the cocycle function. One natural obstruction for the method to carry over is that, in general, area-minimizing hypersurfaces in a given homology class might not be connected, even when the class is primitive.

APPENDIX A. CONNECTEDNESS OF AREA-MINIMIZING HYPERSURFACES ON THE TORUS

Proposition A.1. *Suppose that $3 \leq n \leq 7$ and let g be a Riemannian metric on $\mathbb{T}^n$. Then every homologically area-minimizing hypersurface in a primitive integer homology class is connected and of multiplicity one.*

Proof. Suppose that $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ is a primitive class and let $\Sigma \in \alpha$ be an area-minimizing hypersurface in α , which exists and is smooth by Federer [21]. Assume that $\Sigma = \Sigma_1 + \Sigma_2 + \cdots + \Sigma_k$ and each Σ_i is closed.

Let α_i denote the homology class represented by Σ_i . It follows that $\alpha = \alpha_1 + \cdots + \alpha_k$. Since Σ is area-minimizing, none of the α_i can be zero (otherwise one could remove Σ_i and decrease the mass of Σ while staying inside α), and we have

$$\mathbf{M}(\Sigma) = \sum_{i=1}^k \mathbf{M}(\Sigma_i).$$

Also, by Federer, Σ is embedded. It follows that the distinct components Σ_i are pairwise disjoint. In other words, for $i \neq j$, we have $\Sigma_i \cap \Sigma_j = \emptyset$. Consequently, the homological intersection product $\alpha_i \cdot \alpha_j$ vanishes. On $\mathbb{T}^n$, each α_i can be represented by the projection of an affine plane, and any two affine planes with linearly independent normal vectors intersect. Therefore we must have that α_i and α_j are linearly dependent for all $1 \leq i, j \leq k$.

It follows that there exist $\beta \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ and a positive integer N such that $\alpha = N\beta$. However, since α is primitive, we have $N = 1$, and it follows that Σ has only one connected component of multiplicity one. $\square$

APPENDIX B. MINIMAL FOLIATION IN AN INTEGER HOMOLOGY CLASS

Proposition B.1. *Assume that g is a Riemannian metric on $\mathbb{T}^n$ and $\alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ is a primitive integer class. Then the following are equivalent:*

- (1) *There is a minimal foliation of $\mathbb{T}^n$ by closed, homologically area-minimizing hypersurfaces in α .*
- (2) $\omega(\alpha) = S(\alpha)$.

Proof. If $\mathbb{T}^n$ admits a foliation by a family Φ_t of closed minimal hypersurfaces in α , the following standard calibration argument shows that each leaf Φ_t minimizes the area in the homology class α : consider the following $(n-1)$ -form:

$$w := \iota_\nu \text{vol}_g,$$

where ν is the globally chosen unit normal vector field to the foliation and ι_ν is the interior product.

A standard computation shows that this is a closed calibration form. By Stokes' theorem, for any integral cycle T with $[T] = \alpha$, we have:

$$\mathbf{M}(T) \geq \int_T w = \int_{\Phi_t} w = \mathbf{M}(\Phi_t).$$

Hence, each Φ_t minimizes area among all cycles representing α . Therefore, $\mathbf{M}(\Phi_t) = S(\alpha)$ for all t . It follows that

$$S(\alpha) \leq \omega(\alpha) \leq \sup \mathbf{M}(\Phi(t)) = S(\alpha)$$

implying $\omega(\alpha) = S(\alpha)$.

Conversely, assume that $\omega(\alpha) = S(\alpha)$. By the definition of $\omega(\alpha)$, we can find a sequence of continuous maps $\Phi_i : S^1 \rightarrow \mathcal{Z}_{n-1}(M, \alpha)$ (the space of all cycles in α) such that

$$\lim_{i \rightarrow \infty} \sup_{t \in S^1} \mathbf{M}(\Phi_i(t)) = \omega(\alpha) = S(\alpha).$$

We want to take the limit of the leaves of these sweep-outs. For any geodesic ball $U \subset \mathbb{T}^n$, since Φ_i is a one-parameter sweep-out, there exists $t_i \in S^1$ such that the current $\Phi_i(t_i)$ divides U into two regions of equal volume. Consider the sequence $\{\Phi_i(t_i)\}_i$ of integral cycles with uniformly bounded mass. By compactness, there is a subsequence (still denoted by $\Phi_i(t_i)$) that converges in the flat norm to $\Sigma \in \alpha$. By the lower semicontinuity of mass:

$$S(\alpha) \leq \mathbf{M}(\Sigma) \leq \liminf \mathbf{M}(\Phi_i(t_i)) = S(\alpha).$$

Therefore, Σ is homologically area-minimizing in α . Furthermore, it follows from the flat convergence that Σ also divides U into two parts of equal volume. Hence, the isoperimetric inequality implies that there is a positive constant δ such that $\mathbf{M}(\Sigma \llcorner U) \geq \delta > 0$. Consequently, Σ passes through U . Taking the limit, we see that the union of all homologically area-minimizing hypersurfaces in α is $\mathbb{T}^n$. The fact that it is a minimal foliation follows from a result of Hass in [19], who proved that two area-minimizing hypersurfaces in the same homology class α do not intersect:

Theorem B.2. *Let M^n be an orientable Riemannian manifold. Let G_1, G_2 be hypersurfaces in M , each of which is area-minimizing in its homology class and which are homologous up to orientation. Let F_1 and F_2 be connected components of G_1 and G_2 , respectively. Then either $F_1 \cap F_2$ is empty or F_1 and F_2 coincide.*

□

APPENDIX C. THE UNIFORM CONTINUITY OF THE ENERGY BARRIER

Lemma C.1. *For any metric g on $\mathbb{T}^n$ and any two positive constants $C_1 < C_2$, there exists $K = K(g, C_1, C_2)$ such that for any $C_1 g \leq g', g'' \leq C_2 g$ and $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, we have*

$$|\Delta W_{g'}(r) - \Delta W_{g''}(r)| \leq K|r||g' - g''|_g$$

where $|r|$ denotes the Euclidean norm of the vector $r \in \mathbb{R}^n$ under some identification $H_{n-1}(\mathbb{T}^n, \mathbb{R}) = \mathbb{R}^n$.

Proof. It follows exactly as in [30] that:

$$|\omega_{g'}(r) - \omega_{g''}(r)| \leq \left((1 + C_1^{-1}|g' - g''|_g)^{\frac{n-1}{2}} - 1 \right) C_2^{\frac{n-1}{2}} C \omega_g(r) \quad (\text{C.1})$$

whenever $C_1 g \leq g', g'' \leq C_2 g$. We want to bound $\omega_g(r)$ in terms of $|r|$. Let g_{flat} be the flat metric induced from the Euclidean metric on $\mathbb{T}^n$. Then $g \leq c g_{flat}$ for some $c > 0$. Consider the sweep-out $\Phi(t)$ of $\mathbb{T}^n$ by parallel flat sub-tori with normal direction r . We have

$$\mathbf{M}_{g_{flat}}(\Phi(t)) = S_{g_{flat}}(r) = |r|.$$

It follows that

$$\mathbf{M}_g(\Phi(t)) \leq c^{n-1}|r|, \quad \forall t.$$

Hence, by the definition of the width, we have

$$\omega_g(r) \leq \sup \mathbf{M}_g(\Phi(t)) \leq c^{n-1}|r|. \quad (\text{C.2})$$

Combining (C.1) and (C.2), we have

$$|\omega_{g'}(r) - \omega_{g''}(r)| \leq K_1|r||g' - g''|_g.$$

A similar inequality holds for the stable norm:

$$|S_{g'}(r) - S_{g''}(r)| \leq K_2|r||g' - g''|_g.$$

The desired inequality now follows from the triangle inequality. $\square$

APPENDIX D. A DECAY LEMMA FOR A FINITE-VOLUME GAP

In this section, we prove Lemma 6.3. We first point out that the finite-volume assumption on the gap:

$$\text{Vol}_n(G) < \text{Vol}_n(\mathbb{T}^n, g),$$

gives only a subsequential decay. This means that:

$$\text{Vol}_{n-1}(G \cap \partial B_{R_k}) \rightarrow 0$$

along a suitably chosen sequence of radii $R_k \rightarrow \infty$. To get the full convergence as $R \rightarrow \infty$, we use Harnack's inequality to show that the $(n-1)$ -dimensional area of each spherical slice is controlled from above by the volume of an annulus of fixed width. In the proof below, all the constants C_i depend only on M, N, ρ, r , and the ambient metric.

Proof of Lemma 6.3. Since M and N are area-minimizing, it follows from the standard curvature estimates for stable minimal hypersurfaces that outside of a sufficiently large compact set, one can write M locally as a normal graph over N :

$$M = \{\exp_x(u(x)\nu(x)) : x \in N\}$$

where $u > 0$. Moreover, u satisfies a uniformly elliptic equation with uniformly controlled coefficients. Since G has finite volume, we have that

$$\int_N u d\mathcal{H}^{n-1} < \infty.$$

Assume by contradiction that there exists a positive constant $\epsilon > 0$ together with a sequence $R_k \rightarrow \infty$ such that

$$\text{Vol}_{n-1}(G \cap \partial B_{R_k}) \geq \epsilon, \quad \forall k \in \mathbb{N}.$$

Fix small positive constants ρ and r . For sufficiently large k , $G \cap \partial B_{R_k}$ is contained in the normal tubular region over $N \cap (B_{R_k+r} \setminus B_{R_k-r})$. Cover $N \cap (B_{R_k+r} \setminus B_{R_k-r})$ by intrinsic balls $B_\rho^N(p_j)$ with uniformly bounded overlaps depending on ρ and N . Let $\pi : G \rightarrow N$ be the normal projection outside a sufficiently large compact set. It follows that

$$\text{Vol}_{n-1}(G \cap \partial B_{R_k} \cap \pi^{-1}(B_\rho^N(p_j))) \leq C_1 \rho^{n-2} \sup_{B_\rho^N(p_j)} u. \quad (\text{D.1})$$

Applying Harnack's inequality to u inside each ball $B_\rho^N(p_j)$, we get

$$\int_{B_\rho^N(p_j)} u d\mathcal{H}^{n-1} \geq C_2 \rho^{n-1} \inf_{B_\rho^N(p_j)} u \geq C_3 \rho^{n-1} \sup_{B_\rho^N(p_j)} u. \quad (\text{D.2})$$

Combining (D.1) and (D.2), we have

$$\text{Vol}_{n-1}(G \cap \partial B_{R_k} \cap \pi^{-1}(B_\rho^N(p_j))) \leq C_4 \int_{B_\rho^N(p_j)} u d\mathcal{H}^{n-1}.$$

Summing over j and using the uniformly bounded overlap on the covering:

$$N \cap (B_{R_k+r} \setminus B_{R_k-r}) \subset \bigcup_j B_\rho^N(p_j),$$

we have

$$\epsilon \leq \text{Vol}_{n-1}(G \cap \partial B_{R_k}) \leq C_5 \int_{N \cap (B_{R_k+r} \setminus B_{R_k-r})} u d\mathcal{H}^{n-1}.$$

Passing to a subsequence, we may assume that all $N \cap (B_{R_k+r} \setminus B_{R_k-r})$ are disjoint. It follows that

$$\int_N u d\mathcal{H}^{n-1} \geq \sum_{k=1}^{\infty} \int_{N \cap (B_{R_k+r} \setminus B_{R_k-r})} u d\mathcal{H}^{n-1} \geq \sum_{k=1}^{\infty} C_5^{-1} \epsilon = \infty.$$

This contradicts the fact that $\int_N u d\mathcal{H}^{n-1} < \infty$ and completes the proof. $\square$

APPENDIX E. FINITENESS OF STABLE MINIMAL HYPERSURFACES IN THE GAP

In this section, we prove Proposition 9.5. We first establish that the volume of an open set bounded by a closed, connected, stable minimal hypersurface inside G cannot be arbitrarily small.

Lemma E.1. *There exists a positive constant v_0 depending only on the metric g , such that for any closed, connected, stable minimal hypersurface S inside G , if $S = \partial U_S$, then $\text{Vol}_n(U_S) \geq v_0$.*

Proof. By the standard curvature estimate for stable minimal hypersurfaces [41], there exists a positive constant C such that for all closed stable minimal hypersurfaces S , it holds that

$$\sup_{x \in S} |A_S(x)| \leq C.$$

In particular, the normal curvatures of S are uniformly bounded. It follows from [1], Theorem 1.2, that there exists a positive constant ρ , depending only on C and g , such that

$$\sup_{x \in U_S} d_g(x, S) \geq \rho.$$

In other words, U_S contains a ball of radius ρ . Since the metric is uniformly bounded on $\mathbb{R}^n$, there is a uniformly positive lower bound v_0 on the volume of a ball of radius ρ . We conclude that

$$\text{Vol}_n(U_S) \geq \inf_x \text{Vol}_n(B_g(x, \rho)) := v_0.$$

This proves the lemma. $\square$

We now construct the desired collection of stable hypersurfaces inductively as follows. Start with a closed, connected, stable minimal hypersurface

$$\tilde{S}_1 = \partial U_{\tilde{S}_1} \subset G.$$

By Lemma E.1,

$$\text{Vol}_n(U_{\tilde{S}_1}) \geq v_0 > 0.$$

We claim that there exists a maximal open set U_{S_1} inside G containing $U_{\tilde{S}_1}$ such that S_1 is a closed, connected, stable minimal hypersurface. Suppose not. Then one can find an infinite sequence of nested open sets

$$U_{\tilde{S}_1} \subset U_{\tilde{S}_2} \subset U_{\tilde{S}_3} \subset \dots$$

such that $\partial U_{\tilde{S}_i} = \tilde{S}_i$ is a closed, connected, stable minimal hypersurface.

As in the argument of [46], Lemma 6, there exists $i_0 \in \mathbb{N}$ such that for all $j > i_0$, the hypersurface $\tilde{S}_j$ can be written as a graph locally over $\tilde{S}_{i_0}$, with uniformly bounded slopes. Consequently, $\text{Vol}_{n-1}(\tilde{S}_j)$ is uniformly bounded in terms of $\text{Vol}_{n-1}(\tilde{S}_{i_0})$.

By the compactness theorem for minimal hypersurfaces with uniformly bounded area and curvature, a subsequence of $\tilde{S}_i$ converges smoothly (with possible multiplicity) to a closed minimal hypersurface $S_\infty \subset G$. For sufficiently large k , we can then write $\tilde{S}_k$ as a graph of a smooth function $u_k : S_\infty \rightarrow \mathbb{R}$, where $u_k \rightarrow 0$ in $C^\infty(S_\infty)$. The normalizing sequence

$$v_k := \frac{u_k}{\|u_k\|_{C^{2,\alpha}(S_\infty)}}$$

converges and produces a nontrivial Jacobi field along S_∞ , contradicting the bumpiness of g .

Therefore, such an infinite nested sequence cannot exist, and we have the maximal open set U_{S_1} as claimed. We then remove U_{S_1} from G and repeat the same construction to $G \setminus U_{S_1}$, which still has finite volume. Since we remove an open set of volume at least v_0 at every step and G has finite volume, the process must terminate after finitely many steps, and we obtain an optimal collection $\{S_1, S_2, \dots, S_j\}$ of pairwise disjoint closed, connected, stable minimal hypersurfaces, with the corresponding open sets $\{U_{S_1}, U_{S_2}, \dots, U_{S_j}\}$ such that $S_i = \partial U_{S_i}$.

Claim E.2. *For any closed, minimal hypersurface Σ contained in the interior of $\Omega_{R_k} = G_{R_k} \setminus \left(\cup_{i=1}^j U_{S_i}\right)$, we have*

$$\mathbf{M}(\Sigma) \geq \mathbf{M}(\Gamma_0^{R_k}) + \mathbf{M}(\Gamma_1^{R_k}).$$

Proof of claim. Since $H_{n-1}(\mathbb{R}^n, \mathbb{Z}) = 0$, Σ is separating, we can cut along Σ and consider the component Ω'_{R_k} that contains both $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$.

By the maximality property of the collection $S_1, S_2, \dots, S_j$, the hypersurface Σ is unstable. Hence, it admits a neighborhood foliated by hypersurfaces with strictly smaller area than Σ . Therefore, by minimizing the area in the homology class of Σ inside Ω'_{R_k} , we obtain a stable minimal hypersurface $\tilde{\Sigma}$ such that

$$\mathbf{M}(\tilde{\Sigma}) < \mathbf{M}(\Sigma).$$

In particular, $\tilde{\Sigma} \neq \Sigma$. There are two cases:

Case 1: If $\tilde{\Sigma}$ contains a stable component $\Sigma' \subset \Omega'_{R_k}$. This minimal hypersurface is disjoint from $S_1, S_2, \dots, S_j$. Suppose that inside G we have $\Sigma' = \partial U_{\Sigma'}$. Then $U_{\Sigma'}$ is either disjoint from all the U_{S_i} , or contains some of the U_{S_i} . Nonetheless, both cases contradict the maximality of the collection $\{S_1, S_2, \dots, S_j\}$.

Case 2: If $\tilde{\Sigma}$ contains no closed, stable minimal hypersurfaces in the interior of Ω'_{R_k} , then it must be the whole boundary of Ω'_{R_k} , minus Σ . Since $\Gamma_0^{R_k}$ and $\Gamma_1^{R_k}$ are parts of the boundary of Ω'_{R_k} , we have

$$\mathbf{M}(\tilde{\Sigma}) \geq \mathbf{M}(\Gamma_0^{R_k}) + \mathbf{M}(\Gamma_1^{R_k}).$$

This proves the claim and completes the proof of Proposition 9.5. $\square$

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